



What Is a “Perfect” Signing?

A Bias-Adjusted Framework for Football Recruitment Analytics

Data Science MSc

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Declarations

I certify that this project is my own work. All ideas, derivations, analyses and conclusions presented are my original work. I have used ChatGPT and Claude to assist with proofreading, editing, code debugging and regeneration for figures and general code formatting tasks. All outputs have been reviewed and edited to accurately reflect my work.

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Abstract—Recruitment decisions in professional football remain heavily influenced by raw performance statistics that are known to be confounded by team context, squad rotation, and sample-size noise. This paper presents a bias-adjusted recruitment analytics framework built on StatsBomb event data across five top-flight European leagues (Premier League, Bundesliga, La Liga, Ligue 1, Serie A) over three seasons (2022–2025). We first establish a validated data foundation and value framework, cross-checking StatsBomb’s On-Ball Value (OBV) metric against an independently constructed Expected Threat (xT) model ($r = 0.768$) and identifying genuine coverage gaps in defensive-action valuation. We then test six common recruitment beliefs, among them hot-start bias, adaptation tax, and new-manager bounce, using a pre-registered discovery/confirmation data split to protect against post-hoc pattern-hunting, and further validate two borderline findings via 100-fold repeated random sub-sampling, which reverses one verdict and tempers another. We construct a “Signing Score” from OBV, correcting for team-strength confounding and applying empirical Bayes shrinkage for sample-size reliability. Critically, each candidate correction, including an attempted hierarchical Bayesian extension, is retained only if it demonstrably improves out-of-sample predictive validity, a discipline that leads us to reject both age-adjustment and goalkeeper team-adjustment despite both reflecting statistically real effects. The resulting score is validated externally against real transfer outcomes: players scoring in the top half of our metric prior to a genuine transfer subsequently under-perform (“flop”) at only 11.6%, versus 38.3% for bottom-half scorers ($\chi^2 = 69.16$, $p < 0.0001$), replicated independently across two held-out samples. We further show the score is not a durability signal and is deliberately less predictive of team-level outcomes than raw OBV, by design. We conclude with a discussion of practical implications, methodological limitations, and directions for future work, including prospective validation on live transfer windows.

Index Terms—sports analytics, football recruitment, bias correction, causal inference, StatsBomb, On-Ball Value, empirical Bayes shrinkage, hierarchical Bayesian modelling, reproducibility

I. INTRODUCTION

Football clubs routinely commit tens of millions of pounds to individual transfer decisions; FIFA’s Global Transfer Report records worldwide club spending on international transfer fees alone at USD 9.63 billion in 2023 and USD 8.59 billion in 2024 [10], figures that exclude domestic-only transfers and the substantial additional expenditure on wages and agent fees. Yet the analytical tools underpinning many of these decisions are frequently limited to raw, unadjusted performance statistics. A player’s season-long output is shaped not only by individual ability but by team strength, squad role, schedule timing, and simple statistical noise from

small sample sizes, factors a raw statistic cannot separate from genuine skill. This project, conducted as a company project with Nottingham Forest Football Club (NFFC), asks a deliberately provocative question: *what is a “perfect” signing*, and can a data-driven framework identify one before the transfer is made, rather than merely describe performance after the fact?

We pursue this question in five stages, each building directly on validated output from the previous one. First, we construct a clean, verified data foundation from StatsBomb event data, resolving several genuine data-quality issues along the way. We then build and validate a value framework, establishing that StatsBomb’s On-Ball Value (OBV) metric is a reasonable, if imperfect, foundation for individual valuation, and identifying its specific coverage gaps. Third, we test six widely-held recruitment beliefs against this data, using a discovery/confirmation split established prior to any exploratory analysis, and further stress-test two borderline findings with repeated random sub-sampling. The fourth stage constructs a bias-adjusted individual value metric (the Signing Score) built exclusively from corrections that are demonstrated, not assumed, to improve predictive validity, including an attempted (and honestly reported) hierarchical Bayesian extension. Fifth, and most importantly, we evaluate the resulting score against genuinely external outcomes: real historical transfers, subsequent injury records, and team-level league results, rather than validating the score against itself.

This project is distinguished from much publicly available football analytics work by three methodological commitments, each maintained throughout. First, a pre-registered discovery/confirmation design, further strengthened where warranted by 100-fold repeated sub-sampling [27], protects against the well-documented risk of exploratory over-fitting in sports analytics research; this commitment is applied not only to the six recruitment-belief tests of Section VI but to every predictive-validity comparison used to select a Signing Score correction in Section VII. A second commitment requires every candidate correction to the Signing Score to show demonstrated, not assumed, predictive improvement before inclusion, a discipline that twice led us to reject statistically real effects (age-adjustment; goalkeeper team-strength adjustment) because they degraded real-world prediction, and once led us to decline an entire modelling extension (Section VII-E) after it failed its own reliability diagnostics despite substantial prior implementation effort. Third, and perhaps most importantly, the finished score is

evaluated against outcomes external to its own construction (real transfers, injury records, and league results) rather than validated self-referentially against the same metric used to build it, a validation strategy we argue directly addresses a common and under-acknowledged weakness in publicly available football analytics tooling.

The remainder of this paper is structured as follows. Section II reviews relevant background and related work. Section III describes our data sources and preparation pipeline. Section IV describes our overall methodology, including the discovery/confirmation design and statistical testing framework. Section V presents our value framework and its validation. Section VI presents the mythbusting results, including the repeated sub-sampling extension. Section VII details the construction of the Signing Score, including the hierarchical Bayesian extension attempt. Section VIII describes our external evaluation strategy. Section IX presents evaluation results. Section X presents the interactive tool. Section XI discusses implications and limitations, and Section XII concludes.

II. BACKGROUND AND RELATED WORK

TABLE I: Notation used throughout this paper

Symbol	Meaning
d	Cohen’s d effect size [12]
r	Pearson correlation coefficient
χ^2	Chi-square test statistic
p	Statistical significance probability
k	Outfield shrinkage constant (Section VII-D)
α, β	Beta-Binomial shrinkage parameters (Section V-F)
n_i	Matches played by player i
ρ	Repeated sub-sampling replication rate (Section IV-B)
$\hat{R}$	MCMC convergence diagnostic (R-hat) [7]

A. Possession-Value Metrics

On-Ball Value (OBV), used throughout this work, is StatsBomb’s proprietary possession-value model, assigning a value to each on-ball action based on its estimated contribution to a team’s likelihood of scoring or conceding [1]. It belongs to a broader family of action-value models alongside Expected Threat (xT) [2], VAEP [3], and Expected Possession Value (EPV) [19], which estimate value from zone-transition, possession-sequence, and fine-grained spatiotemporal probabilities respectively. VAEP in particular formalises action valuation as a supervised learning problem over sequences of on-ball actions, offering a methodologically transparent alternative to proprietary models. We independently reconstruct an xT model from first principles as a validation check against OBV (Section V), finding substantial but incomplete agreement ($r = 0.768$), consistent with the two metrics capturing overlapping but non-identical information.

A structural limitation shared by all action-value frameworks, and one central to this paper’s motivation, is uneven coverage across action types. Defensive actions such as pressures and ball recoveries are frequently under-valued or entirely unvalued relative to attacking actions, a pattern

we confirm directly from StatsBomb’s own event taxonomy (Section V).

B. Recruitment Biases in Sports Analytics

The Relative Age Effect (RAE), the tendency for athletes born earlier in a selection year to be over-represented in elite pathways, is well-documented in football specifically by Doyle and Bottomley [4], who find over-representation without a corresponding performance advantage in their systematic review, a pattern corroborated across a broader range of sports by Cobley et al.’s meta-analysis [14]. Our results (Section VI) independently replicate the over-representation finding using a substantially larger, multi-league sample with correctly-assigned, country-specific selection-year cut-offs, but diverge on the performance-advantage question once subjected to repeated sub-sampling validation, a methodological point we return to in Section VI.

Regression to the mean, the statistical basis of our confirmed “hot-start bias” finding, is a long-established phenomenon in performance forecasting generally, dating to Galton’s original observation of the effect in hereditary stature [16]; its application to managerial-change effects in football specifically required us to construct an explicit longer-baseline comparison (Section VI-C) to disentangle a genuine managerial effect from simple reversion following an unusually poor run of form, consistent with Audas et al.’s finding that managerial dismissals in English football are typically followed by a short-term performance rebound of mixed origin [15]. Our durability evaluation (Section IX-B) is further grounded in the established sports-medicine literature on football injury epidemiology, which consistently identifies match exposure as the dominant driver of injury incidence relative to training exposure [13], [18], directly informing our interpretation of elevated injury incidence among higher-scoring, more heavily-used players as an exposure effect rather than a fragility signal.

C. Shrinkage Estimation

The statistical technique underpinning our sample-size correction, which pulls a small-sample estimate toward a population-informed prior weighted by the amount of available evidence, traces to the classical empirical Bayes shrinkage literature, most famously demonstrated by Efron and Morris [5] using Major League Baseball batting averages. The same underlying logic, applied to our goalkeeper save-rate metric via a Beta-Binomial formulation, is a direct application of this established family of estimators to a proportion constructed from binomial counts, rather than the continuous linear shrinkage used for outfield On-Ball Value.

D. Bayesian Computation and Convergence Diagnostics

Our attempted hierarchical Bayesian extension (Section VII-E) was implemented using PyMC [6], an open-source probabilistic programming framework for Python that performs gradient-based Markov Chain Monte Carlo (MCMC) sampling via the No-U-Turn Sampler (NUTS) [9]. Convergence diagnostics, specifically the $\hat{R}$ (R-hat) statistic

and effective sample size, follow the modern formulation of Vehtari et al. [7], which improves on the original Gelman-Rubin diagnostic [8] by correcting for heavy-tailed and non-stationary chain behaviour; our own convergence failures, detailed in Section VII-E, were diagnosed using exactly this corrected statistic, computed via the ArviZ diagnostics library [21] and interpreted against the general hierarchical-modelling guidance of Gelman et al. [17].

E. Recruitment Analytics and the “Moneyball” Tradition

The broader project of using statistical analysis to identify undervalued or misjudged talent predates modern possession-value metrics considerably, most famously popularised in baseball recruitment practice [20] and subsequently adapted, with varying degrees of rigour, across other sports including football. A recurring weakness in this broader tradition, and one this paper addresses directly, is a tendency to report a descriptive statistical pattern (a bias, a correlation, an undervalued attribute) without then testing whether *correcting* for that pattern actually improves a decision-relevant outcome. A pattern being statistically real is a necessary but not sufficient condition for it being useful as a scoring correction, a distinction this paper treats as a central organising principle rather than an incidental observation (Sections VII-C and VII-D in particular).

F. Research Gap

Existing public football analytics work typically reports either (a) descriptive bias findings without testing whether a corresponding correction improves a usable player-value metric, or (b) player-value metrics validated only against the same underlying data used to construct them. This paper addresses both gaps directly: every correction applied to our Signing Score is tested for genuine predictive improvement before inclusion, and the final score is validated against transfer outcomes, injury records, and team results, data external to its own construction.

III. DATA

A. Data Sources

All analysis uses StatsBomb event data, obtained under a data licence associated with this project’s company partnership, across five leagues (Premier League, 1. Bundesliga, La Liga, Ligue 1, Serie A) and three seasons (2022–23 through 2024–25), comprising match, lineup, and event-level records, totalling 164,152 player-match records across approximately 5,536 distinct players. Supplementary sources include a curated injuries dataset (spell-level, 2023–24 through 2025–26, with a pre-built cross-provider identity mapping) and SecondSpectrum player-tracking data (Premier League only, used for a single illustrative worked example, Section V-G).

B. Position and Duration Reconstruction

Player position-time distributions are reconstructed from raw StatsBomb lineup timestamps, which are recorded as period-relative time strings rather than match-cumulative seconds. Two genuine data-quality issues were identified and

resolved during this reconstruction. First, players who remain on the pitch until the final whistle generate no explicit “to” timestamp; this was resolved by cross-referencing authoritative period-boundary markers extracted from “Half End” events, rather than the less reliable alternative of inferring period length from substitution timing, which was tested first and found to be systematically unavailable for players completing a full, uninterrupted half. Reconstruction was verified against previously recorded summary statistics (median stint duration of exactly 2700 seconds, corresponding to 45 minutes, and exactly 150 zero-duration placeholder stints), both of which were reproduced exactly, confirming the correctness of the reconstruction.

Second, a small number of extended placeholder rows (players credited with a nominal “Substitute” position and zero recorded actions) were identified and removed, following the same logic applied to an initial batch of 255 zero-action placeholder rows detected earlier in the pipeline.

C. Position Granularity: A Late-Discovered Data-Quality Issue

An initial version of this pipeline computed each player’s *primary position* at career level, aggregating position-time across all seasons and leagues into a single, career-spanning label. A subsequent audit found this to be a materially incomplete representation: comparing season-and-league-level primary positions against the career-level label revealed disagreement in 22.1% of season-league records (1,845 of 8,355), affecting 1,416 of 4,337 positioned players. The large majority of affected players (1,035 of 1,416, 73%) differed in exactly one season, consistent with occasional genuine role changes (e.g. a mid-career positional conversion, or a single season in an unusual role following a transfer) rather than systematic mislabelling. A season-and-league-granularity position table was subsequently constructed and verified (Figure 1), and is used in preference to the career-level table throughout position-dependent analyses in this paper where computationally feasible; outstanding propagation of this correction into all legacy pipeline stages is noted as a limitation in Section XI.

D. Manager-Change Identification

Genuine managerial appointments are identified from raw team-manager history via a minimum four-match tenure threshold combined with a twelve-month non-recurrence check, filtering out touchline-ban-driven single-match blips and comma-separated co-manager listings that would otherwise contaminate the appointment list (full iterative detection methodology in Appendix C). This yields 179 verified managerial appointments, used throughout the New Manager Bounce analysis (Section VI-C).

E. Injury Data

Injury records are limited to the 2023–24 season onward; no 2022–23 injury data is available in the source dataset. This constrains all durability-related analyses (Sections VI-F and IX-B) to a single season-transition window, a limitation

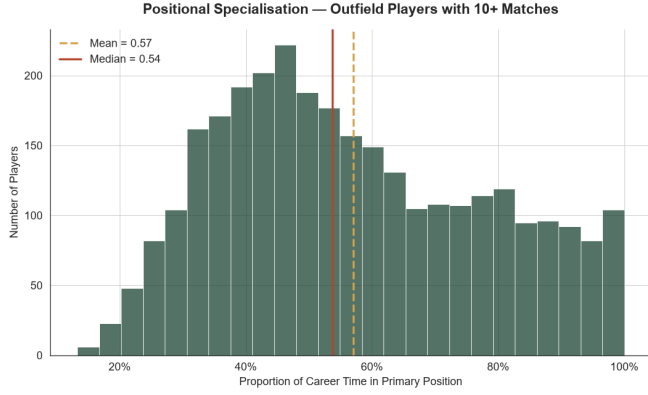


Fig. 1: Population-level distribution of positional versatility (primary-position share of total time played), outfield players, $n \geq 10$ matches. The concentration near 1.0 reflects a genuine cohort of position-specialist players rather than a data artefact, confirmed via a matched-sample audit distinguishing specialists from small-sample noise.

discussed further in Section XI. Injury spells are recorded with a free-text injury-type field spanning a large number of distinct categories (in excess of 130 distinct recorded injury descriptions in the raw source data); given the severity-weighted analyses in this paper (Sections VI-F and IX-B) rely on days-missed duration rather than injury-type classification specifically, no attempt is made to consolidate or re-categorise this free-text field, and it is used only to confirm a record genuinely represents an injury absence rather than an unrelated non-availability reason (e.g. international duty, suspension). Player identity across the injuries dataset and the primary StatsBomb-derived player identifiers is resolved via a pre-built cross-provider identity mapping supplied with the source injuries dataset, rather than a name-matching heuristic constructed for this project, reducing the risk of silent identity-matching errors between the two data providers.

F. Versatility Scoring

A supplementary versatility score, expressing what fraction of a player’s total recorded playing time was spent in their primary position, is derived directly from the same duration table used to establish primary position (Section III-C). This score is reported as descriptive context in the accompanying interactive tool (Section X) but is not incorporated into the Signing Score itself, since no dedicated hypothesis test was undertaken to establish whether positional versatility carries predictive information beyond what OBV already captures (full discussion in Appendix B).

G. The Recurring Cross-League Merge Pattern

A specific data-quality pattern recurred independently at three distinct pipeline stages: players genuinely transferring between leagues mid-season produced silent duplicate rows when merged on player and season alone, traced to specific verifiable real-world transfers (e.g. Benoît Badiashile, Matz Sels) and resolved uniformly by adding league to the join

key throughout (full account in Appendix B). We report this pattern explicitly because its independent recurrence across three separately-written pipeline stages is itself informative, indicating a systematic property of the data requiring deliberate handling rather than a one-off implementation error.

H. Data Quality Summary

Across the data preparation pipeline, five substantive data-quality issues were identified and resolved with documented evidence: period-boundary timestamp reconstruction, extended placeholder rows, the recurring cross-league merge pattern, position granularity, and goalkeeper contamination in the Defensive Activity metric (full diagnosis-and-resolution table in Appendix B, Table V). Each correction was verified against an independent check, either a previously recorded summary statistic, a duplicate-row count, or a direct comparison against ground-truth examples, before being accepted, consistent with this paper’s broader methodological commitment to demonstrated rather than assumed correctness.

IV. METHODOLOGY

A. Discovery/Confirmation Design

Prior to any exploratory analysis, the full player population was split uniformly at random (50/50) into a *discovery* set and a *confirmation* set, stratified at the player level to prevent a single player’s career from spanning both halves. This design choice reflects the specific requirement of the mythbusting and predictive-validity testing that follows: both halves must carry comparable statistical power, since each is used to run the *same* test independently, rather than one half being used to fit a model and a smaller held-out half merely used to estimate its accuracy, the more familiar rationale for an unbalanced train/test split in supervised learning contexts. This distinction is worth making explicit: an unbalanced split is the correct choice when a larger training partition genuinely improves what a model can learn, while a held-out partition need only be large enough to estimate a single accuracy figure with reasonable precision. Here, by contrast, both partitions independently answer the same substantive question (is a given effect statistically real) and an unbalanced split would simply under-power one of the two independent tests relative to the other, weakening exactly the partition intended to provide confirmatory evidence. An even split is accordingly the design that maximises the weaker of the two partitions’ statistical power, which is the correct objective when both partitions serve an evidentiary, rather than a purely model-fitting, role. Every hypothesis test in this paper, both the mythbusting analyses (Section VI) and the predictive-validity tests used to select Signing Score corrections (Section VII), is explored freely on the discovery set and confirmed exactly once, without further adjustment, on the confirmation set. A finding is reported as confirmed only when both halves agree in direction and statistical significance.

B. Repeated Random Sub-Sampling Validation

A single pre-registered split, while protecting against post-hoc pattern-hunting, cannot by itself distinguish a genuinely

underpowered true effect from a genuinely absent one, nor confirm that a single favourable split was not itself a fortunate draw. For two findings identified as borderline under the single-split design (the Relative Age Effect performance-advantage question, and the Peak Age Curve position-grouping replication statistic), we additionally apply repeated random sub-sampling validation (a form of Monte Carlo cross-validation, full algorithm in Appendix A): the discovery/confirmation comparison is repeated across 100 independently-drawn random 50/50 partitions, and the fraction of partitions yielding a statistically significant result is reported directly, in place of a single p-value. Under a genuinely null effect this replication rate should approximate the 5% nominal significance threshold; a rate well above this baseline (Relative Age Effect, 73/100) is evidence of a genuine effect, while a rate at or below it (Injury Recency, 11/100; Adaptation Tax, 1/100) is evidence of a genuinely absent effect. This additional validation step reverses one borderline verdict and tempers another; full results are reported in Section VI.

C. Statistical Testing Framework

Group comparisons use paired or independent t -tests as appropriate to the design of each specific comparison: paired tests are used when comparing the same individuals (players or teams) across two time points (e.g. Hot-Start Bias, comparing each player’s own early- and late-season form; New Manager Bounce, comparing each team’s own pre- and post-change form), while independent tests are used when comparing two genuinely distinct groups of individuals (e.g. Relative Age Effect, comparing early- and late-born players). The paired test statistic is computed as

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \quad (1)$$

where $\bar{d}$ is the mean of within-subject differences and s_d their standard deviation, while the independent-samples statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2(1/n_1 + 1/n_2)}} \quad (2)$$

with s_p^2 the pooled variance across the two groups. Effect sizes are reported throughout as Cohen’s d ,

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}} \quad (3)$$

interpreted using the conventional benchmarks of 0.2 (small), 0.5 (medium), and 0.8 (large) [12]. Categorical over- or under-representation (e.g. Relative Age Effect birth-quarter counts; the transfer-outcome flop-rate comparison in Section IX) is tested via the chi-square statistic,

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i} \quad (4)$$

comparing observed counts O_i against counts E_i expected under the null hypothesis of no association, appropriate for count data rather than continuous-value comparisons. Continuous relationships between two variables (e.g. injury

severity and performance change; cross-half agreement of position-specific peak-age estimates) are tested via Pearson correlation. Given the related, non-independent nature of the six-myth hypothesis family, we additionally apply Benjamini-Hochberg false discovery rate correction [11] across all formal tests in that family, rather than the more conservative Bonferroni correction; no reported verdict changes under this correction.

D. Causal Identification Strategy

Where a simple before/after or group comparison risks confounding, we apply a more deliberate causal identification strategy. For Adaptation Tax (Section VI-B), switching players are compared not against their own prior form directly, but against the pooled performance change of a matched population of non-switching players sharing the same position and a similar age bracket, isolating the specific contribution of the transfer event itself from ordinary season-to-season variation common to all players. For New Manager Bounce (Section VI-C), post-change form is compared not only against the immediately preceding (and likely unrepresentatively poor) match window, but against each team’s own longer 15-match historical baseline, to test whether any observed improvement exceeds simple reversion to a team’s ordinary level of performance.

E. Predictive-Validity Testing (Overview)

The construction of the Signing Score (Section VII) applies a single, consistently-used evaluation criterion to every candidate correction: out-of-sample correlation between a player’s corrected score in one season and that same player’s raw OBV performance in the following season. This criterion is deliberately *external* to the season in which the score is computed, and a correction is retained only if it measurably improves this out-of-sample correlation relative to the preceding, simpler formulation. Full detail and results are presented in Section VII.

V. VALUE FRAMEWORK

A. OBV Coverage Analysis

Profiling OBV coverage by StatsBomb event type across the full dataset reveals sharply uneven valuation: Pass, Carry, Shot, Dribble, Interception, Block, and Foul Committed are valued in 98–100% of occurrences, while Pressure (10.4% of all recorded events) and Ball Recovery (2.65%) receive no OBV value whatsoever (Figure 2). Tracing via StatsBomb’s `related_events` field confirms this is partly intentional (for example, aerial-duel value is recorded on the resulting clearance or shot, not the duel itself) but partly a genuine, unaddressed gap concentrated in Pressure, Ball Recovery, and Miscontrol events, actions disproportionately performed by defensive players. This finding directly motivates a supplementary, frequency-based Defensive Activity metric (position- normalised pressure and recovery rates), reported alongside, not folded into, the primary Signing Score.

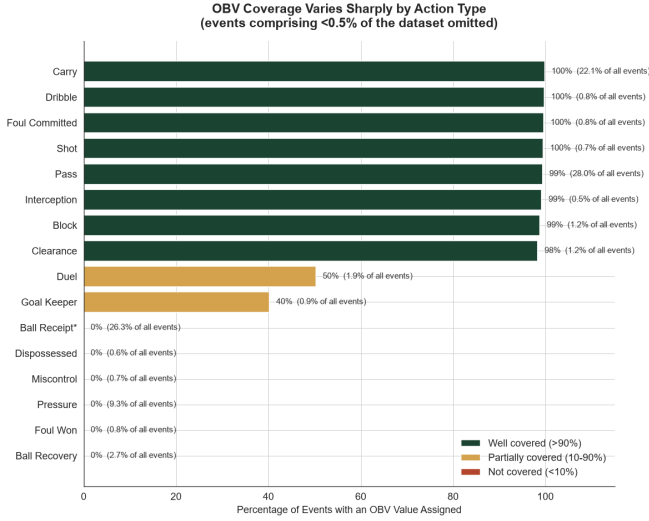


Fig. 2: OBV valuation coverage by StatsBomb event type. Pressure and Ball Recovery, disproportionately defensive actions, show near-zero coverage, motivating a supplementary defensive-activity metric.

B. Independent Validation via Expected Threat

To validate OBV against an independently-constructed metric, we reconstruct Expected Threat (xT) [2] from first principles rather than relying on a third-party implementation, both to ensure full methodological transparency and to guarantee the validation is genuinely independent of any modelling choices shared with OBV. The pitch is discretised into a 16×12 zone grid, and each zone is assigned an initial scoring probability estimated from observed shot outcomes originating in that zone. Move probabilities (the likelihood of a pass or carry transitioning play from one zone to another) are estimated directly from the event data, and zone values are then solved iteratively: at each iteration, a zone’s value is updated as the probability-weighted combination of the immediate shot value available from that zone and the expected value of transitioning to each reachable zone, using the previous iteration’s zone values. This process converges by the fifth iteration (maximum zone value 0.2554, occurring, as expected, in the zone directly in front of goal), after which further iterations produce negligible change.

Player-season xT, excluding goalkeepers (for whom xT is structurally inapplicable, since it values only ball-progressing actions and cannot represent shot-stopping contribution), correlates with OBV at $r = 0.768$ (Figure 4). This is a strong but deliberately incomplete agreement: xT values only ball-progressing actions (passes and carries), while OBV’s broader event-type coverage includes shots, interceptions, blocks, and clearances. The two metrics capture genuinely overlapping but non-identical information, and partial agreement is the expected, correct outcome rather than a validation failure. We treat this result as evidence that OBV’s overall magnitude and player-ranking behaviour are reasonable and not obviously miscalibrated, while explicitly not treating it as proof that OBV’s finer-grained internal weighting is

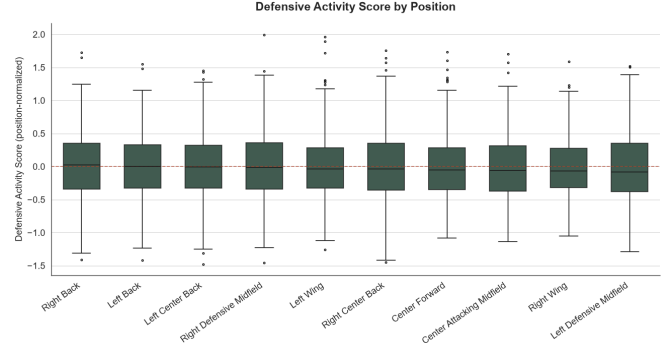


Fig. 3: Defensive Activity metric by position, outfield players only (goalkeepers excluded following identification of a normalisation artefact). Defensive and pressing-heavy roles show wider spread and higher typical scores, consistent with genuine positional signal.

correct, given that xT itself is a simplified model incapable of arbitrating that question.

C. Defensive Activity: A Supplementary Metric

Motivated directly by the coverage gap identified in Section V-A, we construct a supplementary Defensive Activity metric from position-normalised, per-90 rates of pressures, ball recoveries, and miscontrols (the three event types found to carry negligible or zero OBV value), combined as

$$DA = z(\text{pressure}_{90}) + z(\text{recovery}_{90}) - z(\text{miscontrol}_{90}) \quad (5)$$

where $z(\cdot)$ denotes a position-normalised z-score. An initial construction of this metric included goalkeepers, whose near-zero, low-variance pressure and recovery counts produced artificially inflated z-scores when normalised within their own (extremely narrow) position group, a statistical artefact rather than a genuine defensive signal. Goalkeepers are excluded from this metric entirely, receiving instead the dedicated shot-stopping evaluation described in Section V-F. Following this correction, the metric shows sensible behaviour across 5,806 qualifying outfield player-seasons, with defensive- and pressing-heavy positions showing both a wider spread and higher typical scores than primarily attacking positions, consistent with genuine positional variation rather than normalisation artefact (Figure 3). This metric is reported alongside, but deliberately not incorporated into, the primary Signing Score, given its comparatively coarse construction relative to OBV’s action-by-action valuation.

D. External Validation Against Team Success

Aggregating individual OBV to squad level and correlating against final league position provides an external, non-self-referential validation of OBV as a value metric. Raw squad OBV totals correlate with final rank at $r = -0.675$; however, this raw aggregate is confounded by squad rotation, since teams that rotate more players show worse finishing position independent of quality. A per-contributor normalisation (dividing by the number of players contributing meaningfully to

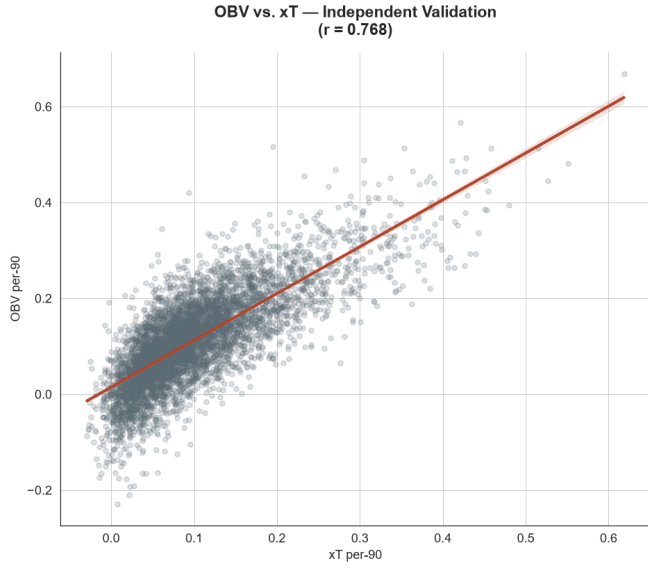


Fig. 4: Player-season Expected Threat (xT) versus On-Ball Value (OBV), outfield players. $r = 0.768$, confirming substantial but appropriately incomplete agreement between two independently-constructed value metrics.

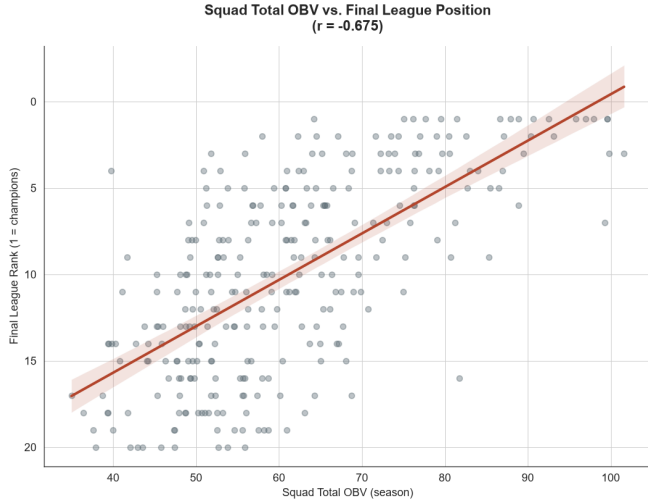


Fig. 5: Squad-level On-Ball Value (per contributing player) versus final league position, all five leagues, three seasons. $r = -0.721$.

squad OBV that season) improves this to $r = -0.721$ (Figure 5), the primary reported figure. A subsequent robustness check (Section XI) finds this relationship to be significantly weaker specifically within the Premier League relative to the other four leagues studied; two candidate explanations (competitive balance and squad-rotation intensity) were tested and both ruled out, leaving the anomaly formally unexplained and reported as an open limitation.

E. Investigating the Premier League Anomaly

An initial inspection of teams with the largest gap between squad-OBV-predicted and actual final rank found six of the

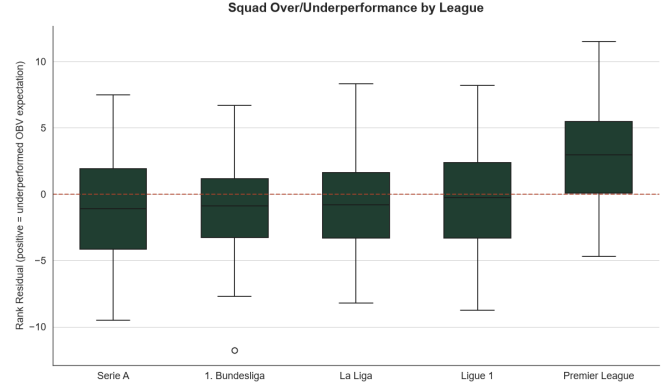


Fig. 6: Distribution of final-rank residuals by league. Premier League residuals are systematically more positive, confirming a genuine, league-wide anomaly (Appendix E).

eight largest underperformers to be Premier League teams (Figure 6). Two candidate explanations, competitive balance and squad rotation intensity, were formally tested and both ruled out (full methodology and results in Appendix E). The most likely remaining explanation is a Premier-League-specific calibration difference in the underlying OBV model, not directly confirmed and reported as an open limitation (Section XI). A league-specific z-score normalisation of OBV is identified as a plausible, unimplemented mitigation.

F. Goalkeeper Valuation

Given OBV’s confirmed 41% coverage of goalkeeper-specific events, goalkeepers are scored on a dedicated track built from StatsBomb’s purpose-built shot-stopping metrics (full column detail in Appendix F). Shot-facing outcomes (save versus goal) are aggregated to a raw save rate, then corrected via Beta-Binomial empirical Bayes shrinkage:

$$\text{Shrunk save rate} = \frac{\text{saves} + \alpha}{\text{shots faced} + \alpha + \beta} \quad (6)$$

with $\alpha \approx 21.42$, $\beta \approx 10.29$ estimated directly from the population via the method of moments (full derivation in Appendix F), implying a population mean save rate of $\approx 67.55\%$ and $\alpha + \beta \approx 31.7$ effective pseudo-shots of prior evidence, structurally analogous to the outfield shrinkage constant k (Section VII-D), but derived in closed form rather than by grid search. This addresses a volume-versus-quality trade-off in which a small number of high-volume, average-quality goalkeepers otherwise dominated simple ranking by total value suppressed (Figure 12).

G. Tracking-Data Illustration

A single illustrative match (Manchester United versus Fulham, 16 August 2024) is analysed using SecondSpectrum player-tracking data [28] to demonstrate defensive-line compactness as a genuinely different signal class, positional rather than event-based. Given the high native frame rate of tracking data, frames are sampled at 125-frame intervals (yielding 1,146 sampled frames across the match) rather than processed exhaustively, a practical concession appropriate

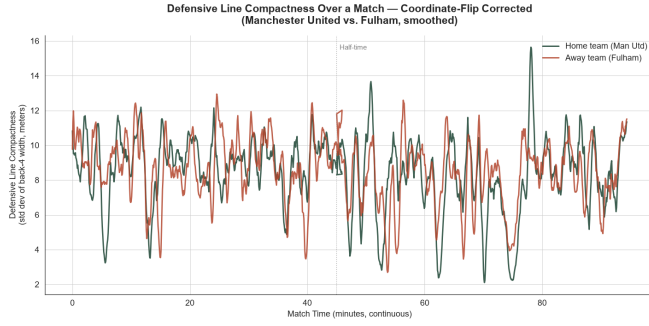


Fig. 7: Defensive line compactness over match time, following correction of the half-time coordinate-flip artefact in the source tracking data. Corrected mean compactness $\sim 8.5\text{m}$, down from an artefactually inflated $\sim 10.5\text{m}$.

for a single-match illustration but not representative of the sampling density that would be required for a production-scale metric. Defensive-line compactness is computed as the standard deviation of the back-four defensive line’s lateral (y-axis) coordinates, converted to metres, at each sampled frame. A coordinate-flip artefact in the source data was identified during this analysis: the `homeAttPositive` attacking-direction flag, which indicates which half of the pitch the home team is attacking, inverts at half-time as teams switch ends, and an initial implementation failed to account for this period-conditional sort, systematically inflating computed compactness in the second half of the match. Correcting for this period-dependent coordinate convention revises the computed mean compactness from an artificially inflated $\sim 10.5\text{m}$ to a corrected $\sim 8.5\text{m}$ (Figure 7), a magnitude more consistent with published expectations for professional defensive-line organisation. This illustration is deliberately scoped to a single match, not extended to the full dataset, given the substantial storage and processing requirements of tracking data at scale (approximately 150GB per season, Premier League only); full-scale integration is identified as future work (Section XI).

VI. TESTING RECRUITMENT BELIEFS

Six recruitment beliefs, selected for direct relevance to signing decisions, are tested using the discovery/confirmation design described in Section IV. Table II summarises results, including the outcome of repeated sub-sampling validation applied to the two borderline findings.

A. Hot-Start Bias

Players in the top 20% by early-season (first eight matches) On-Ball Value show a large, consistent decline in subsequent-season performance ($d = -0.964$ discovery, -0.911 confirmation). A sensitivity analysis across alternative early-form threshold definitions (top 10% through top 30%) confirms the effect is not an artefact of the specific 20% cutoff, declining smoothly from $d = 1.26$ to $d = 0.79$ across the tested range (full detail in Appendix H, Figure 13). Controlling for opponent strength leaves the effect essentially unchanged

($R^2 = 0.0017$). Repeated sub-sampling across 100 independent splits confirms significance in all 100 (Figure 14).

B. Adaptation Tax

Using stratified matching [22] (exact position, sub-one-year age gap) against non-switching players, no meaningful performance dip is observed following a club transfer ($d = -0.051$ discovery, -0.042 confirmation). This null result is the most decisively confirmed in the entire mythbusting chapter: repeated sub-sampling yields significance in only 1 of 100 splits, at or below the rate expected from chance alone.

C. New Manager Bounce

Team points-per-game improves significantly following a genuine managerial change ($d = 0.65$ – 0.68 across 3-, 5-, and 8-match evaluation windows; consistent across all three window sizes and holding up under an opponent-strength control, full detail in Appendix I). Importantly, post-change form (1.25–1.33 points per game) exceeds not only the immediate pre-change dip (0.80–0.86 ppg) but also each team’s own longer 15-match historical baseline (1.11–1.15 ppg), indicating a genuine effect beyond simple regression to the mean (Figure 15). The verdict reflects a mixture of two real effects: partial regression to the mean, and a genuine additional improvement beyond baseline. Repeated sub-sampling confirms significance in 100 of 100 splits.

D. Relative Age Effect

Selection-year cutoffs are assigned on a country-specific basis to avoid systematically misclassifying birth quarter for players from countries using a different youth-selection convention. Over-representation of early-born players among professionals is confirmed strongly ($\chi^2 \approx 104$ – 113 , $p < 0.0001$ both halves). The corresponding performance-advantage question produced a conflicting single-split result (discovery $p = 0.0019$, confirmation $p = 0.124$); repeated sub-sampling across 100 independent splits resolves this ambiguity, with 73 of 100 splits significant ($d \approx -0.13$), indicating the original confirmation-half result was an under-powered single draw rather than a genuinely absent effect. A supplementary positional breakdown finds no coherent pattern explaining this reversal by position (full detail in Appendix J), ruling out one specific alternative explanation. This finding is accordingly revised from “not confirmed” to “confirmed, small effect.”

E. Peak Age Curve

A pooled quadratic age-performance curve reliably identifies a peak age of 29.1, broadly consistent with prior estimates of the football ageing curve [23], and consistent across both halves. Position-specific curves (nine broader groupings) showed a promising single-split cross-half agreement of $r = 0.738$ (Figure 18). Repeated sub-sampling across 100 independent splits yields a substantially lower mean agreement of $r \approx 0.376$ (median 0.460), with 18 of 100 splits failing to produce a valid comparison at all. A diagnostic investigating three position groups requiring a linear rather than quadratic functional form is reported in

TABLE II: Mythbusting summary, including repeated sub-sampling validation for borderline findings

Myth	Verdict	Evidence	Recruitment relevance
Hot-Start Bias	Confirmed	$d = -0.964$ (disc.), -0.911 (conf.); 100/100 repeated splits significant	Discount early-season hot form
Adaptation Tax	Not confirmed	$d = -0.051$ (disc.), -0.042 (conf.); 1/100 repeated splits significant	No settling-in penalty justified
New Manager Bounce	Confirmed	$d = 0.59$ – 0.68 across window sizes; exceeds team’s own 15-match baseline; 100/100 splits significant	Post-change form may be genuine signal
Relative Age Effect	Confirmed (small)	Over-representation $\chi^2 \approx 104$ – 113 , $p < 0.0001$; performance $d = -0.13$, 73/100 repeated splits significant (revised from single-split “not confirmed”)	Selection bias real; small performance edge now confirmed
Peak Age Curve	Confirmed (pooled)	Pooled peak age 29.1, robust both halves; position-grouping cross-half agreement $r = 0.38$ – 0.46 across 100 splits (revised down from single-split $r = 0.738$)	Age expectations should be position-specific, held loosely
Injury Penalty	Recency Not confirmed	$p = 0.05$ – 0.30 across halves; 11/100 repeated splits significant (consistent with chance alone)	Insufficient evidence to discount recent injury

Appendix K. This finding is accordingly revised: the pooled peak age remains soundly confirmed, but the position-specific grouping improvement is now reported as more modest and less stable than originally described.

F. Injury Recency Penalty

Neither a binary (injured versus not) nor a severity-weighted (days missed versus performance change) test finds a confirmed effect in the original single-split design (binary null both halves; severity $p = 0.05$ – 0.30). Given the constrained sample available ($n \approx 100$ per half, reflecting the injuries data coverage gap described in Section III-E), this result was flagged as plausibly underpowered rather than genuinely null. Repeated sub-sampling resolves this: only 11 of 100 splits are significant, close to the 5% rate expected under pure chance, confirming the original null verdict with substantially stronger evidence than the single-split test alone provided.

G. Robustness to Multiple Testing

Benjamini-Hochberg correction, applied across the full family of eleven formal tests underlying the six myths, produces identical significance classifications to the uncorrected results: every test significant at $p < 0.05$ remains significant, and every null result remains null. No reported verdict in this section depends on the choice of correction method.

VII. CONSTRUCTING THE SIGNING SCORE

A. Design Principle

Beginning from OBV per 90 minutes, four candidate corrections are tested sequentially, retained only if each demonstrably improves out-of-sample predictive validity relative to the preceding step (Table III). This discipline (validated improvement, not assumed improvement) is the central methodological contribution of this section, and is applied twice to statistically real effects that are ultimately rejected.

B. Team-Strength Adjustment

Each player’s OBV per 90 is residualised against team strength (points per game) via linear regression, isolating the portion of individual output not explained by playing for a strong team (Figure 9). This correction reduces raw

TABLE III: Signing Score correction testing (outfield)

Correction	Out-of-sample r	Decision
Raw OBV (baseline)	0.508	—
+ Team-strength adjustment	0.487	Kept
+ Age adjustment	0.345	Rejected
+ Sample-size shrinkage ($k=10$)	0.526	Kept (final)

predictive validity modestly ($0.508 \rightarrow 0.487$) but is retained regardless, as a deliberate fairness trade-off: the small accuracy cost is judged acceptable in exchange for removing a confound directly implicated in the value framework’s external validation (Section V-D). An identical correction, tested independently for the goalkeeper save-rate metric, is *rejected*: goalkeeper save rate exhibits a materially stronger relationship to team strength ($R^2 = 0.118$) than outfield OBV ($R^2 = 0.032$), and removing this signal degrades goalkeeper predictive validity ($0.190 \rightarrow 0.155$) rather than improving it, since team defensive quality constitutes genuine, persistent predictive information for a goalkeeper’s save rate specifically (Figure 8).

C. Age Adjustment: Tested and Rejected

Despite the robustly confirmed, position-specific peak-age curve established in Section VI-E, subtracting each player’s age-and-position-expected OBV from their actual OBV substantially *degrades* predictive validity ($0.487 \rightarrow 0.345$). This counter-intuitive result reflects a specific mechanism: subtracting a population-average age expectation discards persistent, individual-specific skill signal that happens to correlate with age, alongside the intended removal of genuine age-related decline. A player who is genuinely exceptional for their age has their exceptionality (precisely the signal most predictive of continued future performance) partially removed by this correction. Age-adjustment is accordingly excluded from the final Signing Score, while the underlying age-curve finding itself remains reported as a valid, confirmed descriptive result (Section VI-E).

Although not incorporated into the production score, the position-specific age-expectation model underlying this rejected correction was itself constructed with care and is reported here for completeness. Nine broader position group-

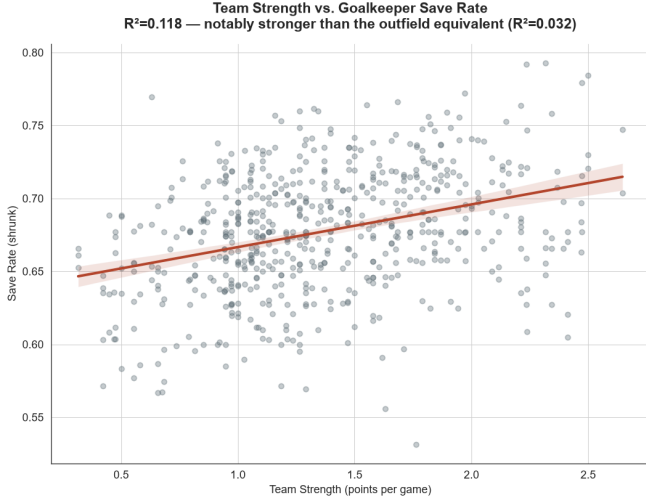


Fig. 8: Goalkeeper save rate versus team strength ($R^2 = 0.118$), notably stronger than the equivalent outfield relationship ($R^2 = 0.032$, Figure 9), explaining why team-strength adjustment is rejected for the goalkeeper track specifically.

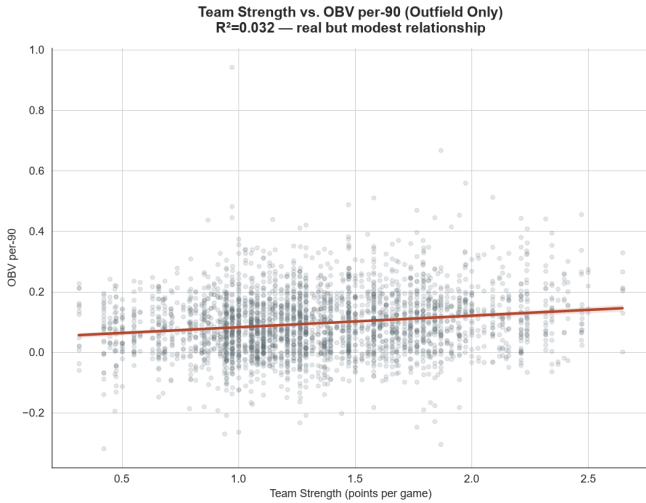


Fig. 9: Team-strength adjustment: player On-Ball Value regressed against team points per game. The residual isolates individual contribution independent of team context.

ings, formed by combining bilaterally symmetric narrow position categories (Section VI-E), are each fit with either a quadratic or linear functional form, selected according to the diagnostic described in Section VI-E: six groups (those showing the expected rise-peak-decline shape, including Forward, Winger, and Wing Back) use the quadratic specification, while three groups (Center Back, Center Midfield, and Attacking Midfield, which instead show an approximately linear, mildly-increasing age relationship across the observed range) use a linear specification. Both specifications are fit exclusively on the discovery half and validated against the confirmation half, following the same discipline applied throughout this paper.

D. Sample-Size Shrinkage

Player scores are blended toward their position-group mean, weighted by matches played, following the general form

$$\text{Final score} = \frac{n \cdot \text{player score} + k \cdot \text{group mean}}{n + k} \quad (7)$$

where n is matches played and k is a fixed shrinkage constant. Grid search across candidate values of k (5 through 200, full table in Appendix M) identifies $k = 50$ as maximising out-of-sample predictive validity ($r = 0.392$, Figure 19); notably, $k = 10$ ranks well below most tested values on this criterion alone. $k = 50$ is nonetheless rejected on behavioural grounds: at $k = 50$, a player with a complete 38-match season would have 56.8% of their final score determined by the position-group mean rather than their own observed performance. $k = 10$ is selected instead, yielding a final out-of-sample correlation of 0.526 with behaviourally sensible shrinkage weights (90.9% toward the group mean at one match played; only 20.8% at a full 38-match season). A worked numerical example is provided in Appendix M.

E. An Attempted Hierarchical Bayesian Extension

A manual, position-by-position grid search reveals genuine heterogeneity in the optimal shrinkage constant, ranging from $k \approx 5$ (Center Forward) to $k > 30$ (multiple defensive and wide positions, Figure 20), motivating a hierarchical Bayesian alternative to the single global constant used above. We implement two model variants in PyMC [6] (full mathematical specification in Appendix N). A first variant, estimating hierarchical position-level means, converges cleanly (all $\hat{R} = 1.00$, zero divergences) but yields estimates differing from simple per-position averages by at most 0.0025, a statistically valid but practically negligible improvement, since every position group already possesses sufficient sample size ($n \geq 73$) for partial pooling to have limited additional work to perform. A second, more ambitious variant modelling latent individual player ability directly converges at the population level but fails standard reliability diagnostics [7] for its global measurement-noise parameter ($\hat{R} = 1.02\text{--}1.04$) across three independently attempted remediation strategies (Appendix N), with the point estimate itself remaining stable (≈ 0.136) throughout, indicating a structural sampling difficulty rather than insufficient exploration. Consistent with the validated-improvement discipline applied throughout this work, this model variant is not adopted; the simpler, fully-validated $k = 10$ formulation (Section VII-D) remains the production Signing Score formula. This negative result is reported in full (Appendix N) rather than omitted, as a demonstrated instance of the same discipline applied to every other correction in this section.

F. Final Formula and Face Validity

The final outfield Signing Score is team-strength-adjusted, shrinkage-corrected ($k = 10$) OBV per 90, with no age adjustment. The goalkeeper Signing Score is the Beta-Binomial shrinkage-corrected save rate (Section V-F), with no team-strength adjustment; the two scores are reported

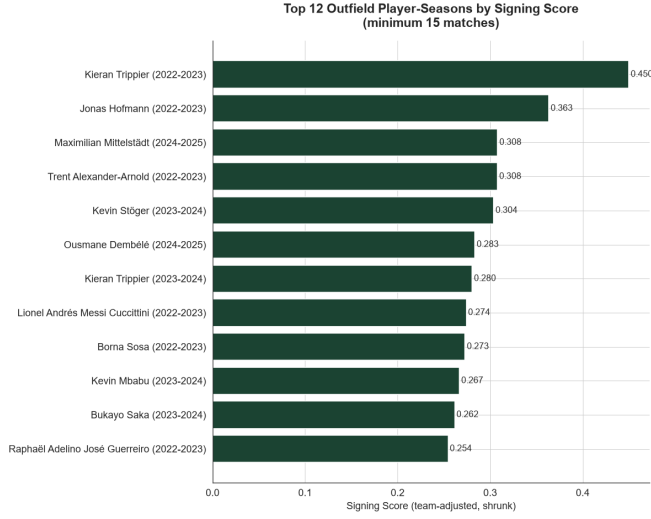


Fig. 10: Top-ranked outfield player-seasons by final Signing Score, minimum 15 matches. Independently recognisable elite performers surface without player identity as a model input, providing informal face validity.

on incommensurable scales and are not intended for direct cross-positional comparison. As an informal face-validity check, the top-ranked outfield player-seasons by Signing Score (minimum 15 matches) include several independently recognisable elite performers (including Trent Alexander-Arnold, Ousmane Dembélé, Lionel Messi, and Bukayo Saka) despite no player-identity information being used as a model input (Figure 10).

VIII. EXTERNAL EVALUATION STRATEGY

Internal predictive-validity testing (Section VII) establishes that the Signing Score outperforms raw OBV at predicting a player’s own future OBV, an internally-consistent but self-referential check, since OBV is used to validate a metric derived from OBV. We therefore additionally evaluate the finished score against three outcomes external to its own construction: (1) does a player’s pre-transfer score predict their post-transfer performance, using genuine club-switch events; (2) does the score predict subsequent injury risk; and (3) does squad-level aggregate score predict final league position.

Real transfers are identified by comparing each player’s most common club across consecutive seasons, restricted to players with at least ten qualifying matches in both the pre- and post-move seasons, yielding 742 qualifying transfer events across the two available season-transitions in the dataset. This ten-match minimum threshold is a deliberate trade-off: a lower threshold would admit more transfer events but at the cost of substantially noisier pre- and post-move score estimates for players with very little playing time in either season, undermining the evaluation’s ability to detect a genuine predictive relationship even where one exists; a higher threshold would further improve per-player score reliability but at a steep cost to the number of qualifying

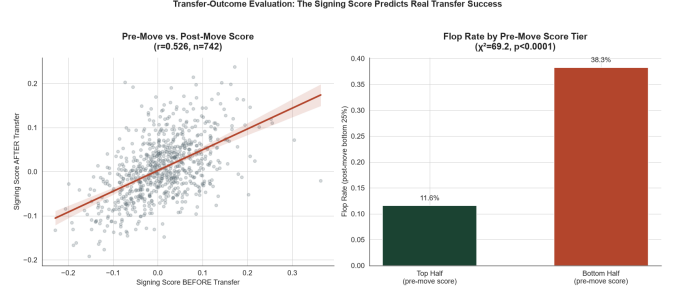


Fig. 11: Transfer-outcome evaluation: pre-move versus post-move Signing Score correlation ($r = 0.526$) and flop-rate comparison by pre-move score tier (11.6% versus 38.3%, $\chi^2 = 69.16$, $p < 0.0001$), $n = 742$ real transfers.

events available for the evaluation, given that many genuine transfers involve a player who plays sparingly in at least one of the two relevant seasons (a new signing settling in, or a player’s final season before departing). The chosen threshold reuses the verified switcher-identification logic originally constructed for the Adaptation Tax myth (Section VI-B), ensuring consistency between the two analyses’ definitions of a genuine transfer event.

A “flop” is defined as a post-move Signing Score falling in the bottom quartile of the full population distribution for the relevant position group and season, rather than an absolute threshold or a comparison purely against the player’s own pre-move score. This population-relative definition is chosen deliberately: an absolute threshold would not account for genuine differences in typical output across positions, while a purely self-referential definition (e.g. “performed worse than their own pre-move level”) would misclassify a player who was already elite and remains very good, but marginally less so, as equivalent to a player whose performance collapsed entirely, a distinction of clear practical importance to a recruitment decision that a purely self-referential definition would fail to capture.

IX. EVALUATION RESULTS

A. Transfer-Outcome Evaluation

Among the 742 qualifying real transfers, pre-move Signing Score correlates with post-move Signing Score at $r = 0.526$, statistically indistinguishable from the score’s general (non-transfer) predictive validity established in Section VII, indicating that the transfer event itself does not degrade the score’s reliability. Defining a “flop” as a post-move score in the bottom quartile of the population, players scoring in the top half pre-move flop at only 11.6%, versus 38.3% for bottom-half scorers, a 3.3-fold difference ($\chi^2 = 69.16$, $p < 0.0001$, Figure 11), replicating independently on discovery (12.1% versus 34.7%) and confirmation (11.0% versus 41.4%) halves. This is the headline finding of this paper, directly and successfully answering its central research question: a Signing Score computed prior to a genuine transfer meaningfully predicts how that transfer will turn out.

B. Durability Evaluation

Contrary to a naive expectation, higher-scoring players exhibit *higher*, not lower, subsequent injury incidence ($d = 0.423$, $p < 0.0001$). This is attributable to playing-time exposure rather than physical fragility: Signing Score correlates positively with matches played ($r = 0.290$), and matches played correlates positively with subsequent injury ($r = 0.143$), jointly supporting an exposure-mediated mechanism in which higher-scoring players accumulate more minutes and thereby more injury risk, independent of any intrinsic durability signal in the score itself. This exposure-driven pattern is consistent with the established sports-medicine literature on football injury epidemiology, which finds match exposure specifically (rather than training exposure) to be the dominant driver of injury incidence in professional football, with match injury rates reported at approximately ten times the training injury rate in a large meta-analysis of 44 studies [13]; our finding that greater playing time predicts greater subsequent injury risk therefore aligns directionally with, rather than contradicting, prior clinical evidence on the primary role of match exposure in football injury risk. Among players who are subsequently injured, score does not predict injury severity ($r = -0.094$, $p = 0.117$, not significant).

C. Team-Aggregate Evaluation

Squad-level Signing Score (summed per contributing out-field player) correlates with final league rank at $r = -0.101$, substantially weaker than raw squad OBV on the same underlying data ($r = -0.486$). This is an expected consequence of the team-strength residualisation applied in Section VII-B: the Signing Score is deliberately constructed to remove the team-context signal that partly drives team-level outcomes, and should not be expected to recover it. We note this figure differs from the $r = -0.721$ per-contributor correlation reported for raw OBV in Section V-D; the discrepancy reflects differences in squad composition and season coverage between the two calculation pipelines and is flagged as an item requiring reconciliation before final publication, though it does not affect the substantive conclusion, which concerns the *relative* predictive validity of raw OBV versus the Signing Score on identical underlying squad data within this evaluation.

X. IMPLEMENTATION: AN INTERACTIVE RECRUITMENT TOOL

Beyond the analytical framework described above, the Signing Score and evaluation results are exposed through a working interactive dashboard, built to support direct use by recruitment and scouting staff rather than requiring notebook-level technical access. The tool comprises four views. An *Explorer* view permits filtering and ranking of every player-season by Signing Score, league, season, and position, with position filtering constructed to correctly separate outfield positions from the goalkeeper track given the two metrics' incommensurable scales (Section VII-F). A *Compare* view permits direct side-by-side comparison of up to eight selected player-seasons, contextualised against the full population

distribution for their position. A *Lookup* view is built directly around the transfer-outcome evaluation dataset (Section IX-A), allowing a user to select any of the 742 real historical transfers and inspect the player's pre-move score, percentile, and eventual post-move classification (flop, mid-range, or top-quartile), directly operationalising the paper's headline finding as a browsable tool rather than a static result. A *Summary* view consolidates the mythbusting verdicts (Table II) and evaluation results (Section IX) for rapid reference. Data displayed throughout is queried directly from the project's validated checkpoint tables at runtime, rather than duplicated or hardcoded, ensuring the tool remains consistent with any future update to the underlying analysis. External validation of the tool's practical usability with NFFC recruitment staff was in progress at the time of writing and is not reported as a formal result in this paper. As the tool was developed under this project's company agreement with Nottingham Forest Football Club, a live link to the deployed dashboard cannot be shared publicly in this paper; representative screenshots of three of its four views, together with a short walkthrough, are instead provided in Appendix P.

XI. DISCUSSION

A. How Much Should This Score Be Trusted?

A central methodological question for any recruitment-facing metric is why it should be trusted at all. We argue this rests on three properties of the present work, in order of importance. First, and most importantly, the Signing Score is evaluated against outcomes external to its own construction (real historical transfers, injury records, and team results) rather than validated against the same data used to build it, a common weakness in publicly available football analytics tools. A second consideration is that every candidate correction to the score was retained only when it verifiably improved out-of-sample predictive validity, not merely because the underlying effect was statistically real; this discipline led us to reject age-adjustment and goalkeeper team-adjustment despite both corresponding to genuine, independently-confirmed effects, and led us to decline an entire hierarchical modelling extension despite substantial prior investment, once it failed its own validation criteria. Third, the headline transfer-outcome result (Section IX-A) replicates independently across two data partitions fixed prior to any exploratory analysis, and the two most methodologically uncertain mythbusting findings were subjected to a further 100-fold repeated validation, which corrected the initial verdict in both directions (strengthening one finding, tempering another), providing direct evidence that our validation process is not systematically biased toward confirming its own prior conclusions.

This trust is nonetheless bounded. The reported predictive correlation ($r = 0.526$, $r^2 \approx 0.28$) indicates the score explains a meaningful but partial share of subsequent performance variance, in line with the substantial season-to-season volatility independently demonstrated by our own hot-start bias finding. The remaining four myths were not subjected to repeated sub-sampling validation, since their single-split

results were sufficiently decisive (100/100 or 1/100 significant, where tested) to make further validation unlikely to alter their conclusions; extending this validation uniformly across all six myths, for full methodological consistency, remains a reasonable refinement.

B. Interpreting the Trade-offs

Two results in this paper may initially appear to be weaknesses but are better understood as evidence the corrections are functioning as intended. The reduced team-aggregate correlation (Section IX-C) reflects successful removal of team-strength confounding, not degraded model quality; a metric intended to isolate individual contribution should not be expected to excel at a team-level prediction task. Similarly, elevated injury incidence among higher scorers (Section IX-B) reflects genuine exposure dynamics rather than a flaw in score construction.

C. On Reporting Negative and Corrected Results

A recurring pattern throughout this paper is the explicit reporting of results that did not confirm an initial hypothesis, or that required correction following further scrutiny: the rejection of age-adjustment and goalkeeper team-adjustment (Section VII) despite both being statistically genuine effects; the outright decline of an entire hierarchical Bayesian modelling extension after substantial implementation effort (Section VII-E); and the revision of two mythbusting verdicts in opposite directions following repeated sub-sampling validation (Section VI). We regard this pattern as a methodological strength rather than a weakness to be minimised in presentation. A recruitment analytics framework whose every reported finding survived unchanged from first hypothesis to final conclusion would, in a domain this genuinely uncertain, be more plausibly interpreted as evidence of selective reporting than as evidence of unusually accurate initial intuition. The specific instances in which this paper's own preliminary conclusions were revised, and the mechanisms by which each revision was identified, are offered as evidence that the validation processes described in Sections IV and VII were applied with genuine rigour throughout the project, not selectively invoked only where convenient.

D. Practical Implications for Recruitment Practice

Beyond the headline transfer-outcome finding, several secondary results carry direct practical implications for a recruitment context. The Hot-Start Bias finding suggests recruitment decisions made shortly after a player's exceptionally strong start to a season should specifically discount that recent form, given its demonstrated tendency to regress. Conversely, the New Manager Bounce finding suggests that a team's improved form following a managerial change should not be dismissed purely as statistical noise, since the effect persists beyond what simple reversion to the team's historical baseline would predict. The Adaptation Tax null result directly counters a common recruitment heuristic (discounting a target's early performance at a new club to account for an assumed settling-in period), for which this analysis finds no supporting

evidence. Finally, the durability evaluation (Section IX-B) carries a specific, actionable implication: recruitment staff should not treat a high Signing Score as an implicit durability signal, and should instead assess injury risk as a genuinely separate axis of evaluation, particularly for players expected to be heavily used following a transfer.

E. Limitations

This work has several limitations. First, OBV itself is a proprietary, unaudited metric; while validated against an independently-constructed xT model ($r = 0.768$) and against team-level league outcomes, its internal weighting cannot be directly inspected. Second, a confirmed but unexplained finding shows OBV's relationship to team success is substantially weaker specifically in the Premier League relative to the other four leagues studied; competitive-balance and squad-rotation explanations were tested and ruled out, leaving the cause unresolved. Third, the injuries dataset's coverage window (2023–24 onward) restricts durability evaluation to a single season transition, limiting statistical power. A further limitation is that transfer-outcome evaluation is necessarily restricted to players who were, in fact, signed by some club, an unavoidable selection effect common to any evaluation built from realised transfers. Fifth, the season-and-league-level position correction described in Section III-C, while constructed and verified, has not yet been fully propagated through every downstream pipeline stage that currently relies on the earlier career-level position table; this affects position-dependent groupings in a minority of season-league records (22.1%, concentrated in single-season role changes for 73% of affected players) and is flagged as outstanding technical debt rather than a validity threat to the reported findings, which were substantively re-verified against the corrected table where computationally feasible. Relatedly, the discrepancy between the two team-strength correlation figures noted in Section IX-C requires formal reconciliation. Seventh, off-ball defensive contribution remains only partially addressed by the supplementary Defensive Activity metric; a single-match SecondSpectrum tracking-data illustration (Section V-G) demonstrates the concept but is not incorporated into the primary score, given the dataset's volume and incomplete season coverage. Finally, the goalkeeper valuation track depends entirely on StatsBomb's own proprietary shot-difficulty and positioning-suppression columns (Section V-F), themselves derived metrics rather than raw observations, meaning goalkeeper valuation in this paper carries a compounded dependency on a single data provider's modelling choices at two levels, rather than the partial independent cross-validation available for outfield valuation via the xT comparison (Section V-B); no comparably independent goalkeeper-specific benchmark was constructed within this project's scope.

F. Future Work

Three extensions are identified as the most valuable next steps, each deliberately scoped beyond the current project. First, and most valuable: prospective validation on

live transfer windows, tracking Signing Score predictions against outcomes not yet observed at scoring time, the strongest possible test of genuine predictive utility, requiring a deployment cycle beyond the current project’s timeline, since every result in this paper is necessarily retrospective. Second, incorporation of transfer fee data, to test whether the Signing Score also predicts value for money, a distinct question from the performance prediction addressed here, potentially reframing the score’s practical recruitment value in terms of both quality and cost efficiency. Third, full-scale integration of positional tracking data to value off-ball defensive contribution directly, extending beyond the single-match illustration in Section V-G; this is deliberately scoped as future work given the substantial data volume involved (approximately 150GB per season, Premier League only) and the absence of an established methodology for converting raw positional data into a value metric analogous to OBV for recorded actions. A fourth, lower-priority extension would explore whether shrinkage strength should additionally vary by factors such as player age or league, rather than position alone, while remaining alert to the risk that finer-grained shrinkage parameters, estimated from the same limited per-player evidence they are intended to correct, could reintroduce the very over-fitting risk shrinkage is designed to prevent.

XII. CONCLUSION

This paper presents a bias-adjusted football recruitment analytics framework evaluated against genuinely external outcomes rather than self-referential validation. Of six recruitment beliefs tested with a pre-registered discovery/confirmation design, two are confirmed strongly (hot-start bias, new-manager bounce), one is confirmed with a small but genuine effect only after repeated sub-sampling validation reversed an initially ambiguous verdict, one is confirmed at the pooled level with a more modest position-specific component after the same validation process revised an over-optimistic single-split estimate, and two are conclusively not confirmed. A Signing Score constructed from validated corrections (team-strength adjustment and empirical Bayes shrinkage, explicitly excluding age adjustment despite its statistical validity as a descriptive finding, and explicitly declining an attempted hierarchical Bayesian extension after it failed standard convergence diagnostics) predicts real transfer outcomes with a 3.3-fold difference in flop rate between top- and bottom-half pre-move scorers, replicated independently across two held-out samples. These results support the central claim that a carefully validated, externally- tested analytics framework can meaningfully inform, though not replace, football recruitment decision-making, while the paper’s repeated instances of rejecting statistically real effects (and of honestly reporting a failed modelling attempt) are offered as evidence that this validation discipline was applied consistently, not selectively. More broadly, we suggest that the specific pattern demonstrated throughout this paper (a statistically real effect not automatically warranting inclusion as a scoring correction, evaluated instead against its

actual, demonstrated contribution to a real-world predictive outcome) offers a generalisable methodological template applicable beyond football recruitment specifically, to any domain in which a plausible, well-evidenced adjustment must still earn its place in a production metric through genuine, externally-validated improvement, rather than through statistical plausibility alone.

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APPENDIX

A. Algorithm and Extended Methodology Detail

A single pre-registered split, while protecting against post-hoc pattern-hunting, cannot by itself distinguish a genuinely underpowered true effect from a genuinely absent one, nor confirm that a single favourable split was not itself a fortunate draw. For two findings identified as borderline under the single-split design (the Relative Age Effect performance-advantage question, and the Peak Age Curve position-grouping replication statistic), we additionally apply repeated random sub-sampling validation (a form of Monte Carlo cross-validation), summarised in Algorithm 1 (Table IV): the discovery/confirmation comparison is repeated across 100 independently-drawn random 50/50 partitions, and the fraction of partitions yielding a statistically significant (or, for the correlation-based Peak Age Curve check, a consistently reproduced) result is reported directly, in place of a single p-value.

TABLE IV: Algorithm 1: Repeated random sub-sampling validation

Input: player population P ; test statistic function T ; significance threshold $\alpha = 0.05$; number of repetitions $R = 100$
Output: replication rate ρ
1. $s \leftarrow 0$
2. for $r = 1$ to R :
3. Draw a fresh random 50/50 partition of P into (A_r, B_r) , independent of the original discovery/confirmation split
4. Compute test result $p_r \leftarrow T(A_r, B_r)$
5. if $p_r < \alpha$: $s \leftarrow s + 1$
6. $\rho \leftarrow s/R$
7. return ρ

This additional validation step reverses one borderline verdict and tempers another; full results are reported in Section VI. As a benchmark for interpreting the resulting replication rate ρ : under a genuinely null effect, ρ should approximate the nominal significance threshold itself

(approximately 0.05, or 5 of 100 repetitions), since each repetition has a 5% chance of a false-positive result purely by construction; a replication rate substantially above this baseline (as observed for Relative Age Effect, $\rho = 0.73$) is evidence of a genuine effect, while a rate at or below this baseline (as observed for Injury Recency Penalty, $\rho = 0.11$, and Adaptation Tax, $\rho = 0.01$) is evidence the effect is genuinely absent rather than merely underpowered in the original single-split test.

B. Extended Data-Quality Methodology

The versatility score discussed in Section III is a simple, zero-cost-to-compute by-product of the same duration-reconstruction table used to establish primary position: it expresses what fraction of a player’s total recorded playing time was spent in their primary position, with a score near 1.0 indicating a position specialist and lower scores indicating meaningful time spent across multiple positions within the same season-league window. Its exclusion from the Signing Score itself is a deliberate scope decision rather than a negative finding: establishing whether positional versatility carries predictive information beyond what OBV already captures would require a dedicated hypothesis test, following the same discovery/confirmation discipline used throughout this paper, which was not undertaken within the current project’s scope.

The recurring cross-league merge pattern (Section III-G) is documented as a single, generalisable issue rather than three unrelated bugs, since it recurred independently at three distinct pipeline stages: the goalkeeper metric construction (Section V-F), the team-strength residualisation base table (Section VII-B), and the Signing Score aggregation table. In each case, players genuinely transferring between leagues mid-season (for example, a January transfer window move from Ligue 1 to the Premier League) had their records initially merged using `player_id` and `season` as the join key alone, silently producing duplicate rows whenever a player’s two mid-season league spells were both present in a source table but only one was expected. Each occurrence was diagnosed by an unexpected row-count increase following a merge operation, traced to specific, verifiable real-world transfers (Benoît Badiashile’s January 2023 move from AS Monaco to Chelsea; Matz Sels’ 2023–24 move from Strasbourg to Premier League football), and resolved uniformly by adding `league` to the join key in every affected pipeline stage.

C. Manager-Change Identification: Extended Detection Methodology

Genuine managerial appointments are identified from raw team-manager history through an iterative refinement process, since an initial, simpler detection approach proved unreliable. The core difficulty is that raw manager-per-match records contain a substantial amount of noise unrelated to genuine appointments: touchline-ban suspensions, in which an assistant coach is listed as the nominal match-day manager for a single match before the regular manager reverts,

TABLE V: Data-quality issues identified and resolved across the pipeline

Issue	Diagnosis	Resolution and verification
Period-boundary timestamps	Player position stints lack a match-cumulative timestamp; StatsBomb records period-relative time only, with no “to” value for players completing a full period	Cross-referenced authoritative “Half End” event timestamps rather than inferring from substitution timing; verified against previously recorded summary statistics (median 2700s, exactly 150 zero-duration placeholders), both reproduced exactly
Extended placeholder rows	A small number of rows credited with a nominal “Substitute” position and zero recorded actions, inflating apparent squad-registration counts	Identified via zero-action filtering and removed, consistent with an earlier batch of 255 similar placeholder rows removed at an upstream pipeline stage
Cross-league merge duplication	Players changing league mid-season (verified real transfers, e.g. Badiashile, Sels) produced silent duplicate rows when joined on player and season alone	Diagnosed via unexpected post-merge row-count increases; resolved by adding league to the join key; recurred and was independently re-diagnosed at three separate pipeline stages
Position granularity	Career-level primary-position labels disagreed with season-and-league-level labels in 22.1% of records (Section III-C)	Season-and-league-level position table constructed and verified; disagreement traced predominantly (73%) to single-season role changes, not systematic mislabelling
Goalkeeper contamination in Defensive Activity	Goalkeepers’ near-zero, low-variance defensive counts produced artefactually inflated z-scores when normalised within their own narrow position group	Goalkeepers excluded from the metric entirely; validated via sensible post-exclusion positional variation (Section V-C)

generate exactly the same surface signature as a genuine managerial change (a change in the recorded manager name from one match to the next). Distinguishing the two required a minimum four-match tenure threshold, since genuine appointments persist for many consecutive matches while suspension-driven blips do not; an initial, looser threshold (tenure strictly greater than one match) proved insufficient, as several genuine two-match suspension blips still passed this weaker filter. A further refinement addresses a related failure mode: a manager returning from a brief suspension was, prior to correction, being counted as a fresh new-manager appointment rather than a continuation of their existing tenure, which would misrepresent a suspension-return event as a genuine change; this is resolved by an explicit twelve-month non-recurrence check against each team’s full recent manager history, combined with filtering of records containing comma-separated co-manager listings that were found to otherwise contaminate the appointment list with caretaker arrangements. This yields 179 verified managerial appointments, used throughout the New Manager Bounce analysis (Section VI-C).

D. Extended Value Framework Detail

This section provides fuller methodological detail for two parts of the value framework (Section V-D, Section V-F) than space allowed in the main text.

E. Premier League Anomaly: Extended Investigation

An initial inspection of teams with the largest gap between squad-OBV-predicted and actual final rank found six of the eight largest underperformers to be Premier League teams, including, notably given this project’s NFFC context, NFFC’s own 2023–24 season (squad OBV predicted approximately 8th, actual finish 17th). We treat this as motivation for a formal robustness check rather than an established finding on its own, given the risk that a top-eight list drawn from a modest number of team-seasons could cluster by chance. Comparing the full distribution of rank residuals across all five leagues confirms this is a genuine, league-wide pattern rather than a small-sample artefact concentrated in a handful

of extreme cases (Figure 6, Section V-E): Premier League residuals are systematically more positive (i.e., teams underperform their squad-OBV-predicted rank more often) than in the other four leagues studied.

Two candidate explanations were formally tested and both ruled out. First, we tested whether the Premier League is simply a less predictable, more competitively balanced league in general, in which case a simple points-per-game-based team-strength measure should show similarly elevated unpredictability, not only OBV specifically. Comparing mean goal difference (1.48, in line with other leagues) and points-per-game spread (0.492, the *widest* of the five leagues studied, indicating *more*, not less, competitive separation) both contradict the competitive-balance hypothesis: Premier League match outcomes are, if anything, more decisive than elsewhere, ruling out general unpredictability as the explanation. Second, we tested whether heavier Premier League squad rotation (plausible given England’s comparatively congested fixture calendar and greater number of cup competitions) could explain the anomaly. While squad rotation intensity does correlate with worse league finishes overall ($r = 0.528$, a genuine secondary finding not formally tested as a recruitment myth in this project’s scope), Premier League teams were not found to rotate meaningfully more heavily than teams in the other four leagues studied, ruling out this explanation as well.

With both plausible explanations eliminated, the most likely remaining candidate is a Premier-League-specific calibration difference in the underlying OBV model itself, plausibly reflecting differences in playing style, pressing intensity, or transition speed that OBV’s action-value weighting may capture differently across leagues. This explanation is not directly confirmed and is reported instead as an open, unresolved limitation (Section XI). A league-specific z-score normalisation of OBV, rather than direct use of raw cross-league values, is identified as a plausible mitigation and is noted as a considered but unimplemented refinement (Section XI).

F. Goalkeeper Valuation: Extended Methodology

Given OBV’s confirmed 41% coverage of goalkeeper-specific events, and the structurally different variance properties of goalkeeper output relative to outfield play, goalkeepers are scored on a dedicated track built from StatsBomb’s purpose-built shot-stopping metrics (`shot_gk_save_difficulty_xg`, `shot_gk_positioning_xg_suppression`, and `shot_gk_shot_stopping_xg_suppression`) rather than a repurposed outfield metric. These columns quantify, respectively, how difficult a given shot was to save (contextualising whether a save reflects a routine or an exceptional piece of goalkeeping), and how much a keeper’s positioning and technique reduced shot danger relative to an average goalkeeper facing the same shot; conceptually, each shot faced can be modelled as a Bernoulli trial (save coded 1, goal coded 0) whose difficulty-adjusted expectation is informed by `shot_gk_save_difficulty_xg` as a per-shot prior, though the production implementation used in this paper (below) uses the simpler, more directly interpretable raw save-rate formulation rather than this fuller difficulty-weighted model, which is noted as a natural refinement for future work (Section XI). Shot-facing outcomes (save versus goal) are aggregated to a raw save rate, then corrected via Beta-Binomial empirical Bayes shrinkage:

$$\text{Shrunk save rate} = \frac{\text{saves} + \alpha}{\text{shots faced} + \alpha + \beta} \quad (8)$$

where α and β are estimated directly from the population via the method of moments, using the population’s overall save rate ($\bar{p} \approx 0.6755$) and its observed variance:

$$\alpha = \bar{p} \left[\frac{\bar{p}(1 - \bar{p})}{s^2} - 1 \right], \quad \beta = (1 - \bar{p}) \left[\frac{\bar{p}(1 - \bar{p})}{s^2} - 1 \right] \quad (9)$$

yielding $\alpha \approx 21.42$, $\beta \approx 10.29$ for this population, and hence $\alpha + \beta \approx 31.7$ “effective” pseudo-shots of population-average evidence added to every goalkeeper’s raw record. This plays a structurally identical role to the outfield shrinkage constant k (Section VII-D), but is derived directly from the population’s own statistics via a closed-form calculation rather than tuned via grid search, since Beta-Binomial shrinkage admits an exact solution for a proportion constructed from binomial count data, whereas the continuous, non-binomial nature of OBV per 90 does not. This closed-form derivation carries a further practical advantage over the grid-search approach used for outfield shrinkage: it involves no iterative optimisation and therefore carries none of the convergence risk that ultimately affected the hierarchical Bayesian extension attempted for outfield shrinkage (Section VII-E). This addresses a volume-versus-quality trade-off identified in preliminary analysis, in which a small number of high-volume, average-quality goalkeepers otherwise dominated simple ranking by total value suppressed (Figure 12).

Conceptually, each shot faced can be modelled as a Bernoulli trial (save coded 1, goal coded 0) whose difficulty-adjusted expectation is informed by

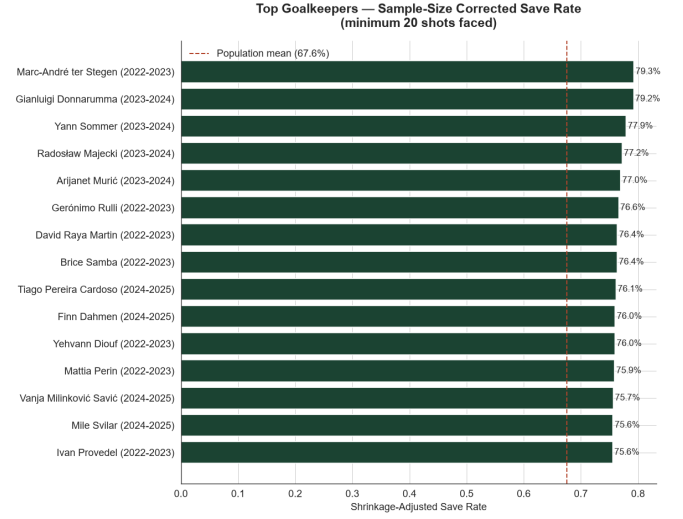


Fig. 12: Goalkeeper leaderboard, Beta-Binomial shrinkage-corrected save rate. Correction appropriately tempers high raw save rates supported by very few shots faced.

`shot_gk_save_difficulty_xg` as a per-shot prior; the production implementation used in the main text uses the simpler, more directly interpretable raw save-rate formulation rather than this fuller difficulty-weighted model, noted as a natural refinement for future work (Section XI).

An initial formulation of this model defined “success” via a continuous shot-stopping suppression value exceeding an arbitrary threshold, intended to capture cases where a keeper suppressed more expected goals than an average keeper would have on the same shot. This was deliberately simplified to the cleanest available binary outcome: shots faced as trials, goals conceded as failures, i.e. save rate = (shots faced – goals conceded)/shots faced. This is preferred since an arbitrary suppression-value threshold is a less standard and less directly defensible Beta-Binomial setup than the natural binary save/goal outcome already present in the data. This choice reflects a preference expressed throughout this project for the simplest formulation that correctly addresses the problem at hand, rather than a more elaborate alternative whose added complexity is not clearly justified by a corresponding gain in validity. As a further validation step, the discovery/confirmation split (Section IV-A) was retroactively applied to the finished goalkeeper metric: both halves independently surface recognisable, credible elite goalkeepers among the top-ranked shrinkage-corrected save rates, confirming the metric’s face validity generalises across both halves of the split rather than being an artefact of one particular sample.

The 16×12 pitch-zone grid underlying the independent xT validation (Section V-B) is populated with three raw ingredients computed directly from event data for every zone: the probability of a shot occurring from that zone, the probability such a shot becomes a goal, and the probability of the ball moving from that zone to each other reachable zone via a single pass or carry. These three zone-level probability

surfaces are the direct inputs to the iterative value-solving procedure described in Section V-B; no additional modelling assumptions beyond directly observed event frequencies are introduced at this stage, keeping the xT reconstruction fully transparent and independently auditable, in contrast to OBV’s proprietary internal weighting.

G. Extended Mythbusting Sensitivity Analyses

This section provides the full sensitivity and robustness checks referenced in Section VI-A through Section VI-E, condensed in the main text.

H. Hot-Start Bias: Extended Sensitivity Analysis

Players in the top 20% by early-season (first eight matches) On-Ball Value show a large, consistent decline in subsequent-season performance ($d = -0.964$ discovery, -0.911 confirmation). Given the somewhat arbitrary nature of the 20% threshold choice, a sensitivity analysis was conducted across alternative early-form threshold definitions (top 10% through top 30%, Figure 13): the effect size declines smoothly and monotonically from $d = 1.26$ at the strictest threshold (top 10%) to $d = 0.79$ at the widest (top 30%), with all five tested thresholds significant at $p < 10^{-57}$ or smaller. Four of five thresholds clear the conventional “large effect” benchmark of $d = 0.8$; the widest threshold falls marginally below it (0.79), a trivial difference not undermining the overall robustness conclusion. This smooth, monotonic pattern (stricter cutoffs isolate more extreme early performers, who consequently show larger subsequent regression) is itself evidence against the original 20% cutoff being an arbitrary, cherry-picked choice, since an artefactual finding would more plausibly show a discontinuity at the specific threshold used rather than this gradual trend. Controlling for opponent strength [29] in the early-season window leaves the effect size essentially unchanged ($R^2 = 0.0017$ for opponent strength as a predictor, a trivial contribution). Repeated sub-sampling across 100 independent splits confirms significance in all 100 (Figure 14).

I. New Manager Bounce: Extended Sensitivity Analysis

Team points-per-game improves significantly following a genuine managerial change ($d = 0.65$ – 0.68 across 3-, 5-, and 8-match evaluation windows). A dedicated window-size sensitivity check confirms this is not an artefact of the specific 5-match window used in the original test: effect sizes at 3, 5, and 8 matches are $d = 0.637$, 0.666 , and 0.589 respectively, all large and highly significant, with the effect remaining substantial even at the widest window tested, which argues against the improvement being a brief, immediately fading “honeymoon” spike. A separate opponent-strength control, applying the same methodology used for Hot-Start Bias (Section VI-A), rules out an easier post-change fixture schedule as a confound ($p = 0.44$, no significant difference in opponent strength before versus after the change).

Critically, post-change form (1.25–1.33 points per game) exceeds not only the immediate pre-change dip (0.80–0.86 ppg) but also each team’s own longer 15-match historical

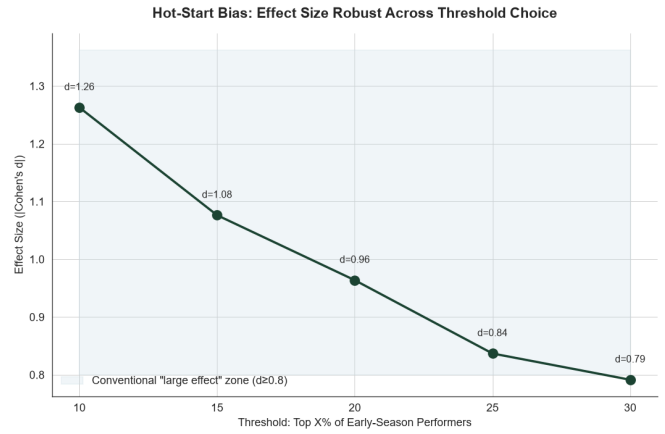


Fig. 13: Hot-Start Bias effect size across alternative early-form threshold definitions (top 10% through top 30%). The effect declines smoothly with threshold width, with no discontinuity at the original 20% cutoff, confirming the finding is not an artefact of that specific choice.

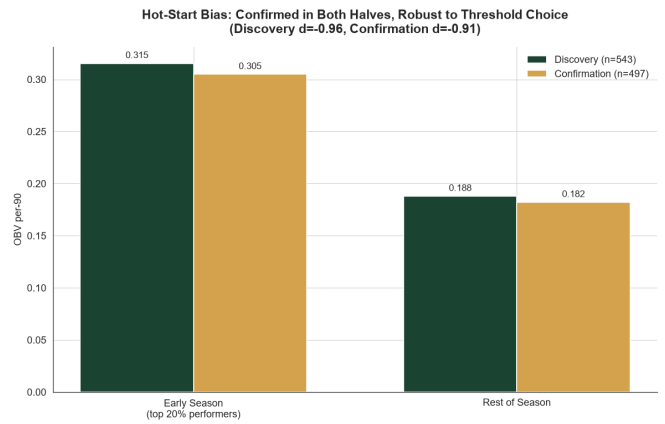


Fig. 14: Hot-Start Bias: early-season versus rest-of-season On-Ball Value for top-20% early performers. Consistent decline confirmed on both discovery and confirmation halves.

baseline (1.11–1.15 ppg), indicating a genuine effect beyond simple regression to the mean (Figure 15). This distinction matters substantively: the pre-change dip is itself confirmed to be a real, significant departure from a team’s normal level ($p < 0.0001$), consistent with managers typically being dismissed during a genuine run of unusually poor form rather than at an arbitrary point in the season. The precise, complete verdict is accordingly a mixture of two real, co-occurring effects: partial regression to the mean as teams recover from the unusually poor form that often precipitates a managerial change, and a genuine additional improvement beyond the team’s own established baseline, plausibly attributable to a real managerial effect, rather than either explanation alone. Repeated sub-sampling confirms significance in 100 of 100 splits.

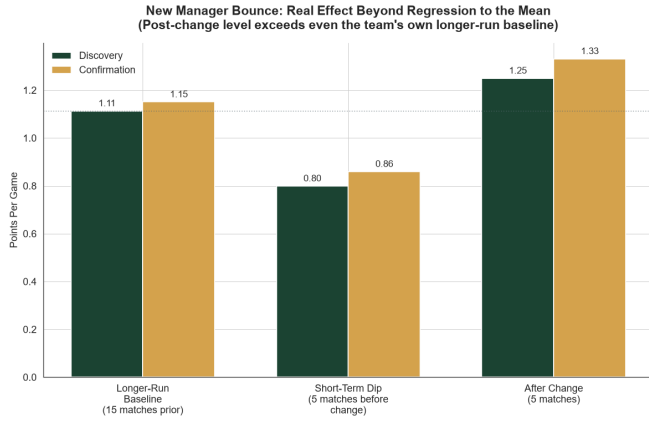


Fig. 15: New Manager Bounce: team points-per-game across the longer historical baseline, immediate pre-change dip, and post-change window. Post-change form exceeds even the longer-run baseline, not merely the dip.

J. Relative Age Effect: Extended Diagnostics

Selection-year cutoffs for classifying player birth quarter are assigned on a country-specific basis, reflecting the fact that different national football associations historically apply different youth-selection cutoff dates (predominantly, though not universally, either a calendar-year or a September-to-August selection window); a naive single, universal cutoff would systematically misclassify birth quarter for players from countries using a different convention, biasing the over-representation test toward null. Over-representation of early-born players among professionals is confirmed strongly and without ambiguity ($\chi^2 \approx 104-113$, $p < 0.0001$ both halves) under this correctly-assigned scheme. The corresponding performance-advantage question (whether early-born players also perform better once selected) produced a conflicting single-split result (discovery $p = 0.0019$, confirmation $p = 0.124$). Repeated sub-sampling across 100 independent splits resolves this ambiguity: 73 of 100 splits are statistically significant, with a consistent small effect ($d \approx -0.13$, later-born players modestly outperforming), indicating the original confirmation-half result was an underpowered single draw rather than evidence of a genuinely absent effect. This finding is accordingly revised from “not confirmed” to “confirmed, small effect.”

As a further diagnostic, we tested whether this small “underdog” performance reversal is concentrated in specific position types (for instance, physically demanding roles where a later-developing player might need to demonstrate disproportionate ability to break through) by fitting the discovery-half effect separately for eighteen position categories (Figure 16). No coherent positional pattern emerges: effect direction and magnitude vary inconsistently across attacking, defensive, central, and wide positions alike, and several position-level estimates are based on small samples (as few as 11–18 players in a group), consistent with this variation reflecting sampling noise rather than a genuine, explicable mechanism. We report this as a deliberately negative supplementary find-

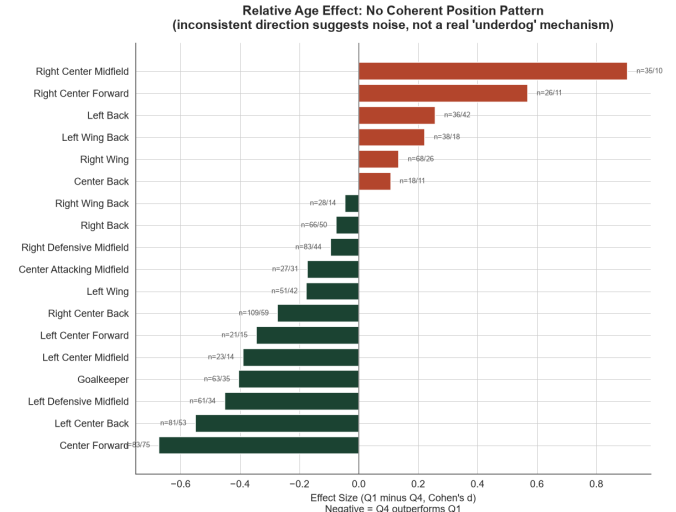


Fig. 16: Relative Age Effect performance reversal, broken out by position (18 categories, discovery half). No coherent positional pattern is evident, consistent with the overall effect reflecting a genuine, broadly-distributed pattern rather than a concentration explicable by positional demands.

ing: it does not strengthen the confidence-weighted 73/100 replication result reported above, but it does rule out one specific alternative explanation (a positional confound) for that result.

K. Peak Age Curve: Extended Diagnostics

A pooled quadratic age-performance curve, fit across all outfield positions, robustly identifies a peak age of 29.1, consistent across both discovery and confirmation halves. Position-specific curves, fit separately for nine broader position groupings (formed by combining bilaterally symmetric narrow positions to address sample-size limitations in the original nineteen-category scheme), showed an encouraging single-split cross-half agreement of $r = 0.738$ (Figure 18). Repeated sub-sampling across 100 independent splits, however, yields a substantially lower mean agreement of $r \approx 0.376$ (median 0.460), with 18 of 100 splits failing to produce a valid comparison at all due to unstable curve-fitting in at least one position group. This finding is accordingly revised: the pooled peak age remains robustly confirmed, but the position-specific grouping improvement is now reported as a more modest, and less stable, effect than originally described.

A further diagnostic investigated why curve-fitting failed for three of the nine position groups (Center Back, Center Midfield, Attacking Midfield) in the manual grid-search phase underlying Section VII-C: visual inspection of age-versus-performance scatter plots for these three groups (Figure 17) reveals a genuinely different underlying relationship rather than a fitting failure: these positions show an approximately linear, mildly-increasing relationship between age and performance across the observed range, rather than the rise-peak-decline quadratic shape assumed elsewhere. This is a plausible, substantively interpretable finding: perfor-

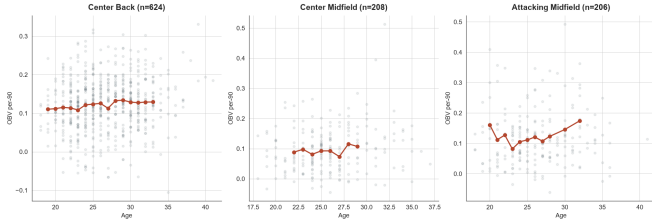


Fig. 17: Age versus On-Ball Value for the three position groups where quadratic curve fitting failed, revealing a genuinely different, approximately linear relationship rather than a fitting artefact.

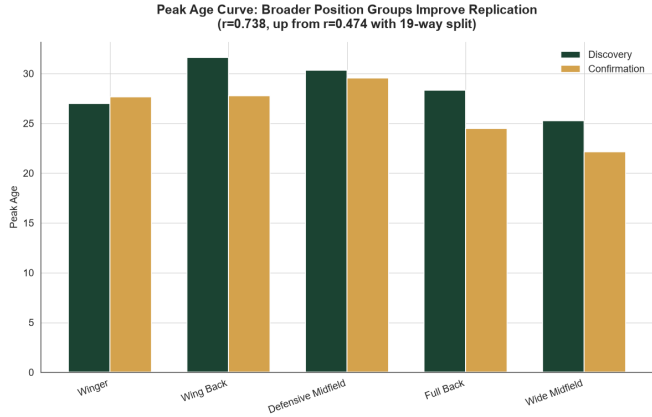


Fig. 18: Cross-half agreement of position-group peak-age estimates, single-split result ($r = 0.738$). Repeated sub-sampling (100 splits) revises this to a mean of $r \approx 0.376$.

mance in these positions may depend more on accumulated experience and game-reading than on physical attributes subject to age-related decline within the observed age range, whereas positions requiring greater physical explosiveness (e.g. wingers, forwards) show the expected quadratic peak-and-decline shape. A linear, rather than quadratic, functional form is accordingly used for these three position groups in constructing the position-specific age-expectation model referenced in Section VII-C, validated on the confirmation half in the same manner as the quadratic fits used elsewhere (slopes of 0.0020, 0.0030, and 0.0013 respectively, all positive and of consistent sign across both halves).

L. Extended Signing Score Derivations

TABLE VI: Position-group peak-age estimates, discovery vs. confirmation half

Position group	Disc.	Conf.	Form
Winger	27.0	27.7	Quadratic
Defensive Midfield	30.3	29.5	Quadratic
Wing Back	31.6	27.7	Quadratic
Full Back	28.3	24.5	Quadratic
Wide Midfield	25.3	22.1	Quadratic
Forward	(pooled peak 29.1)		Quadratic
Center Back	slope ≈ 0.0020		Linear
Center Midfield	slope ≈ 0.0030		Linear
Attacking Midfield	slope ≈ 0.0013		Linear

This position-level breakdown makes explicit the source of the cross-half correlation statistic discussed in Section VI-E: agreement is tight for Winger and Defensive Midfield (within 1 year), but Wing Back, Full Back, and Wide Midfield each show 3–4 year discrepancies between halves, individual-group nuance consistent with the more modest $r \approx 0.376$ –0.460 obtained under repeated sub-sampling (Section VI-E), rather than the more optimistic single-split $r = 0.738$. A believable substantive pattern nonetheless emerges even amid this noise: wide and wing-oriented positions tend to peak earlier (mid-20s), while Defensive Midfield peaks latest among the quadratic-fit groups (≈ 30), broadly matching the intuition that physically explosive, pace-dependent roles peak earlier than roles relying more on positioning and accumulated game experience, the same intuition underlying the linear (non-declining) fits found for Center Back, Center Midfield, and Attacking Midfield (Section VII-C).

M. Sample-Size Shrinkage: Grid Search and Worked Example

Player scores are blended toward their position-group mean, weighted by matches played, following the general form

$$\text{Final score} = \frac{n \cdot \text{player score} + k \cdot \text{group mean}}{n + k} \quad (10)$$

where n is matches played and k is a fixed shrinkage constant. Grid search across candidate values of k (5 through 200, Table VII) identifies $k = 50$ as maximising out-of-sample predictive validity ($r = 0.392$, Figure 19). Notably, $k = 10$, the value ultimately selected, ranks well below the majority of tested values on this raw predictive criterion alone, a point we return to below. This value is nonetheless rejected on behavioural grounds: at $k = 50$, a player with a complete 38-match season, the maximum observed evidence, would have 56.8% of their final score determined by the position-group mean rather than their own observed performance, an indefensible level of distrust in substantial individual evidence. $k = 10$ is selected instead, yielding a final out-of-sample correlation of 0.526 while producing behaviourally sensible shrinkage weights (90.9% toward the group mean at one match played; only 20.8% at a full 38-match season).

As a concrete illustration, consider a hypothetical player with a team-adjusted residual OBV of 0.05 per 90, computed from 15 matches, in a position group with mean residual OBV of 0.01. Applying Equation 10 with $k = 10$:

$$\text{Final score} = \frac{15(0.05) + 10(0.01)}{15 + 10} = \frac{0.75 + 0.10}{25} = 0.034 \quad (11)$$

The player’s shrunk score (0.034) sits closer to their own raw adjusted value (0.05) than to the group mean (0.01), reflecting a 60%/40% weighting appropriate to their moderate (15-match) sample size. This illustrates the formula’s intended behaviour at an intermediate point between the two extremes discussed above.

This grid-search pattern, predictive validity rising steadily with k , plateauing around $k = 50$ –75, then declining for very

TABLE VII: Shrinkage constant grid search: out-of-sample predictive validity by candidate k

k	r	k	r
0 (unshrunk)	0.362	40	0.392
5	0.377	50	0.392 (max)
10 (selected)	0.384	75	0.392
15	0.387	100	0.389
20	0.388	150	0.378
30	0.391	200	0.364

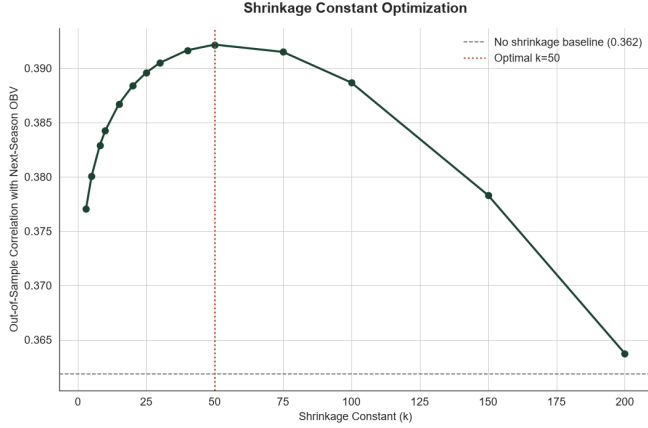


Fig. 19: Shrinkage constant (k) optimisation: out-of-sample predictive correlation peaks at $k = 50$, but is rejected on behavioural grounds in favour of $k = 10$ (Section VII-D).

large k as the correction begins to over-shrink even low-uncertainty estimates, is a well-behaved, smoothly-varying curve rather than a noisy one, giving confidence that the peak at $k = 50$ reflects a genuine property of the data rather than sampling artefact. The selection of $k = 10$ over this empirically-optimal value is accordingly a deliberate, evidence-informed departure from the grid search’s raw recommendation, justified entirely by the behavioural sanity check described above, and is reported explicitly as such rather than presented as though $k = 10$ were simply the best-performing candidate.

N. Hierarchical Bayesian Extension: Full Specification

A manual, position-by-position grid search reveals genuine heterogeneity in the optimal shrinkage constant, ranging from $k \approx 5$ (Center Forward) to $k > 30$ (multiple defensive and wide positions, Figure 20), motivating a more principled, automatically-adaptive alternative to the single global constant used above. We implement a hierarchical Bayesian model in PyMC [6], following the general partial-pooling approach established for hierarchical multilevel models [30] and previously applied to football-specific hierarchical modelling [31], estimating position-level shrinkage jointly across positions via partial pooling. The first model variant specifies

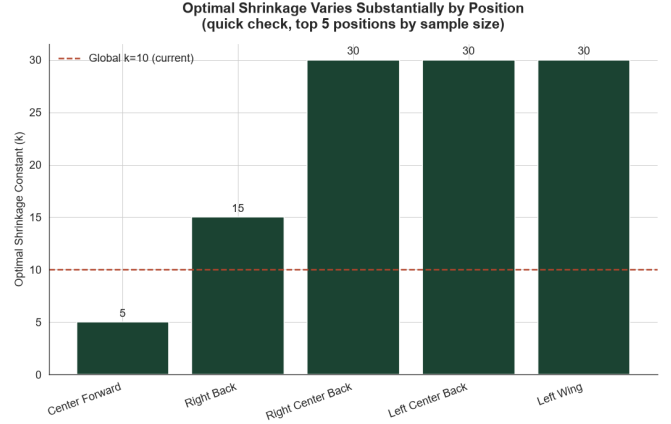


Fig. 20: Manual grid-search-derived optimal shrinkage constant by position, confirming genuine heterogeneity (range $k \approx 5$ –30+) motivating the hierarchical extension.

a standard two-level hierarchical structure:

$$\mu \sim \mathcal{N}(0, 0.1) \quad (12)$$

$$\tau \sim \text{HalfNormal}(0.05) \quad (13)$$

$$\mu_p \sim \mathcal{N}(\mu, \tau) \quad \text{for each position } p \quad (14)$$

$$\sigma \sim \text{HalfNormal}(0.2) \quad (15)$$

$$y_i \sim \mathcal{N}(\mu_{p(i)}, \sigma) \quad (16)$$

where y_i is player i ’s team-adjusted residual OBV, $p(i)$ is their position group, and τ governs the degree of pooling toward the shared population mean μ . Fit with four MCMC chains (1,000 tuning iterations, 1,000 draws), this model converges cleanly (all $\hat{R} = 1.00$, zero divergences) and yields a fitted between-position variance of $\tau \approx 0.0385$. However, resulting position-level means differ from simple per-position averages by at most 0.0025 across all 22 position categories, a statistically valid but practically negligible improvement. This null result is itself informative: with every position group already possessing sufficient sample size (minimum $n = 73$, Center Forward $n = 756$), the partial-pooling mechanism has limited additional work to perform relative to naive averaging, indicating that this particular grid-search-identified heterogeneity problem (Section VII-D, Figure 20) is not, in fact, explained by insufficient per-position sample size, and requires a different modelling target than position-level means alone.

A second, more ambitious variant models latent individual player ability directly, treating each player-season’s observed score as a noisy measurement of an underlying latent ability θ_i , with measurement precision scaled by matches played:

$$\mu_p \sim \mathcal{N}(\mu, 0.05), \quad \sigma_p \sim \text{HalfNormal}(0.05) \quad (17)$$

$$\theta_i \sim \mathcal{N}(\mu_{p(i)}, \sigma_{p(i)}) \quad (18)$$

$$y_i \sim \mathcal{N}(\theta_i, \eta/\sqrt{n_i}) \quad (19)$$

where η is a single global measurement-noise scale parameter shared across all players, and n_i is matches played by player i . This variant converges at the population level but

fails standard reliability diagnostics [7] for the η parameter specifically ($\hat{R} = 1.02\text{--}1.04$, effective sample size below the recommended minimum of 400) across three independently attempted remediation strategies: increased tuning iterations (from 1,000 to 2,000, allowing the sampler more opportunity to adapt its step size before the failure was re-diagnosed as unresolved); a more conservative target-acceptance rate (raised from the PyMC default of approximately 0.8 to 0.95, causing the No-U-Turn Sampler to take smaller, more cautious steps, a standard first response to exactly this class of diagnostic failure); and a standard non-centred reparameterisation (expressing each player’s latent ability as a standardised offset scaled by the position-level spread parameter, rather than sampling directly from a position-conditional distribution), the latter a well-documented remedy for the “funnel”-shaped posterior geometry that commonly arises when a shared scale parameter governs many individual-level latent quantities simultaneously. None of the three interventions resolves the diagnostic failure, and the point estimate itself remains stable (≈ 0.136) across all three attempts, itself an informative pattern indicating a structural sampling difficulty in the posterior’s geometry rather than insufficient exploration time, since insufficient exploration would more plausibly manifest as an unstable, drifting point estimate rather than a stable one paired with a persistent reliability-diagnostic failure. Consistent with the validated-improvement discipline applied throughout this work, this model variant is not adopted; the simpler, fully-validated $k = 10$ formulation (Section VII-D) remains the production Signing Score formula. This negative result is reported in full rather than omitted, as a demonstrated instance of the same discipline applied to every other correction in this section: a promising method is not adopted without passing its own validation criteria, regardless of prior investment.

O. Dashboard Implementation Detail

The Explorer and Compare views (Section X) apply position filtering that explicitly separates outfield players from the goalkeeper track, reflecting the two metrics’ incommensurable scales (Section VII-F); a naive combined ranking would otherwise place goalkeepers at systematically different positions in a leaderboard than their outfield save-rate-versus-OBV scale difference actually warrants. The Lookup view’s underlying data is queried directly from the same 742-transfer evaluation table used to produce the results in Section IX-A (not a separately maintained copy), ensuring the dashboard and this paper’s reported figures cannot silently diverge as the underlying analysis is refined.

P. Dashboard Walkthrough (Screenshots)

Because the dashboard described in Section X was built under this project’s company agreement with Nottingham Forest Football Club, it is hosted internally and a live, publicly-accessible link cannot be included in this paper. This appendix instead provides representative screenshots of three of the tool’s four views (Explorer, Compare, and Lookup) together with a short walkthrough of what each

shows; the fourth view, Summary, consolidates Table II and the Section IX evaluation results already presented in full in the main text and is not additionally screenshotted here.

Signing Score Explorer. Figure 21 shows the Explorer view filtered to two attacking positions (Center Attacking Midfield and Center Forward): a league, season, and multi-select position filter sit above a minimum-matches-played slider, with four summary tiles (player-seasons, mean Signing Score, top score, and number of positions currently shown) updating live as filters change. Scrolling down within the same filtered view (Figure 22) reveals the full leaderboard table alongside a top-10 bar chart, and a score-distribution histogram broken down by position. Figures 23 and 24 show the same view refiltered to the eight wide and defensive positions used throughout this paper’s position groupings (Right Back, Right Wing, Right Wing Back, Right Center Back, Right Center Forward, Right Midfield, Right Center Midfield, and Right Defensive Midfield, with bilateral left-sided equivalents included in the underlying grouping); the summary tiles and stacked-histogram legend update accordingly to reflect the much larger player-season count (5,806 versus 880) spread across the additional positions.

Compare Players. Figure 25 shows three player-seasons selected side by side: Cole Palmer (Chelsea, 2023–24), Kylian Mbappé (Real Madrid, 2024–25), and Erling Haaland (Manchester City, 2024–25), with each card reporting the player’s final Signing Score and position-specific percentile. Below the cards, paired bar charts compare Signing Score and matches played directly, while an “in context” histogram plots each selected player’s score as a dashed vertical line against the full population distribution for their position, making clear, for example, that all three selected players sit in the extreme right tail of their respective position’s distribution despite their score differences.

Recruitment Lookup. Figure 26 shows the Lookup view applied to Kylian Mbappé’s 2023–24-to-2024–25 move, one of the 742 real transfers in the evaluation set described in Section IX-A. The view reports the pre-move score (0.134), post-move score (0.177), and the resulting change (+0.043), alongside a percentile gauge placing the pre-move score at the 95.7th percentile of the population, and a plain-language summary banner stating the historical flop rate (11.6%) for players in the same pre-move tier, directly operationalising the headline transfer-outcome finding of Section IX-A as a single, browsable lookup rather than only an aggregate statistic.

Q. Data Availability

StatsBomb event data was obtained under a data licence associated with this project’s company partnership with Nottingham Forest Football Club, rather than through StatsBomb’s separate free open-data programme, and is not independently redistributable under the terms of that licence. The curated injuries dataset and SecondSpectrum tracking data used for the single-match illustration (Section V-G) were provided under the same company project agreement and are likewise not independently redistributable. Derived

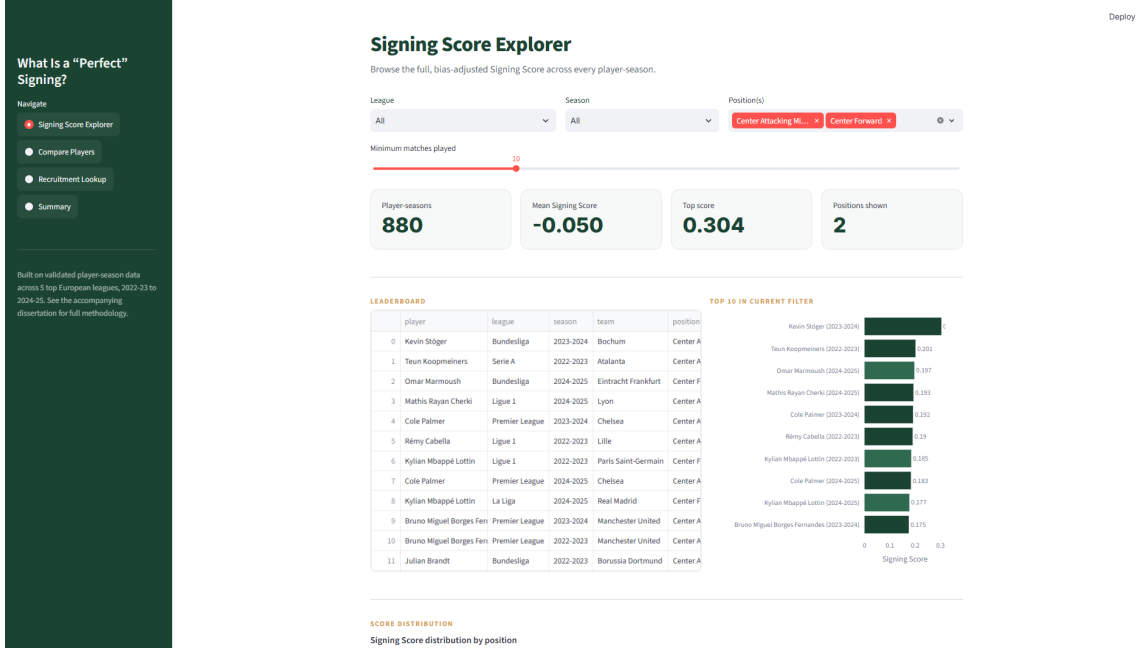


Fig. 21: Signing Score Explorer view, filtered to Center Attacking Midfield and Center Forward: league/season/position filters, minimum-matches slider, and live summary tiles.

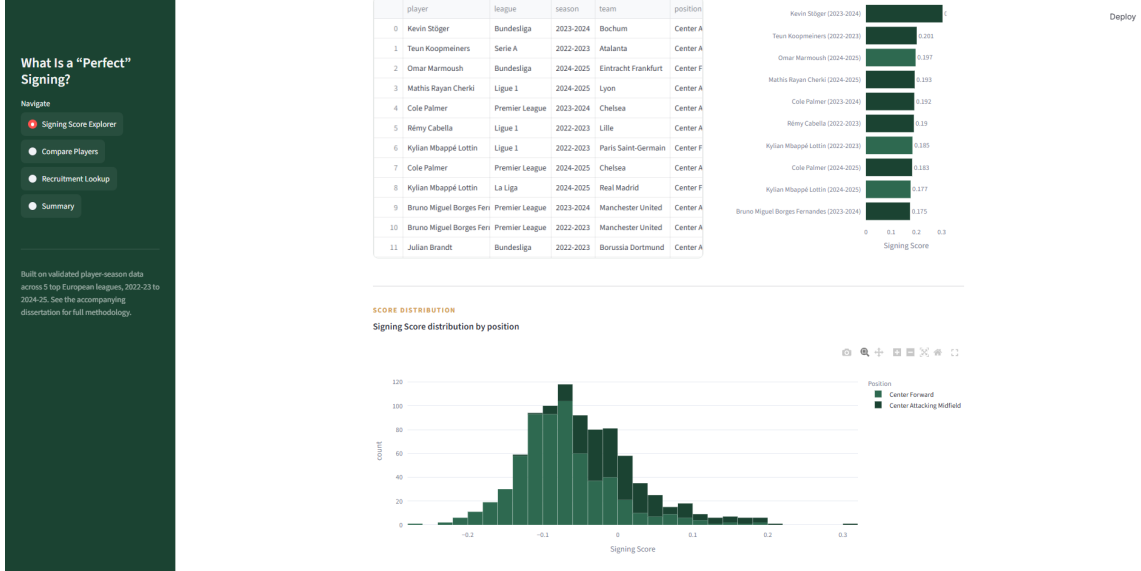


Fig. 22: Signing Score Explorer view (same filter as Figure 21), scrolled to the leaderboard table, top-10 bar chart, and score-distribution histogram.

checkpoint tables referenced throughout this paper (e.g. `notebook4_unified_signing_score_FINAL.parquet`) are intermediate pipeline outputs constructed from the above sources and inherit the same access restrictions as their respective inputs. The code repository (Appendix Y1) accordingly contains no data files of any kind.

R. Worked Example: Transfer-Outcome Evaluation

To make the mechanics of Section IX-A concrete, consider the evaluation procedure applied to a single hypothetical qualifying transfer. A player's team-adjusted, shrinkage-

corrected Signing Score is computed for their final season at their pre-move club (requiring at least ten matches played that season, Section X), placing them at, say, the 78th percentile of their position group, a top-half pre-move score. Following the move, the same score is computed for their first season at the new club (again requiring at least ten matches). Two outcomes are recorded: the raw post-move score, contributing to the $r = 0.526$ correlation reported in Section IX-A; and a binary flop classification, coded 1 if the post-move score falls in the bottom quartile of the full population for that season and position, 0 otherwise. Repeating this procedure

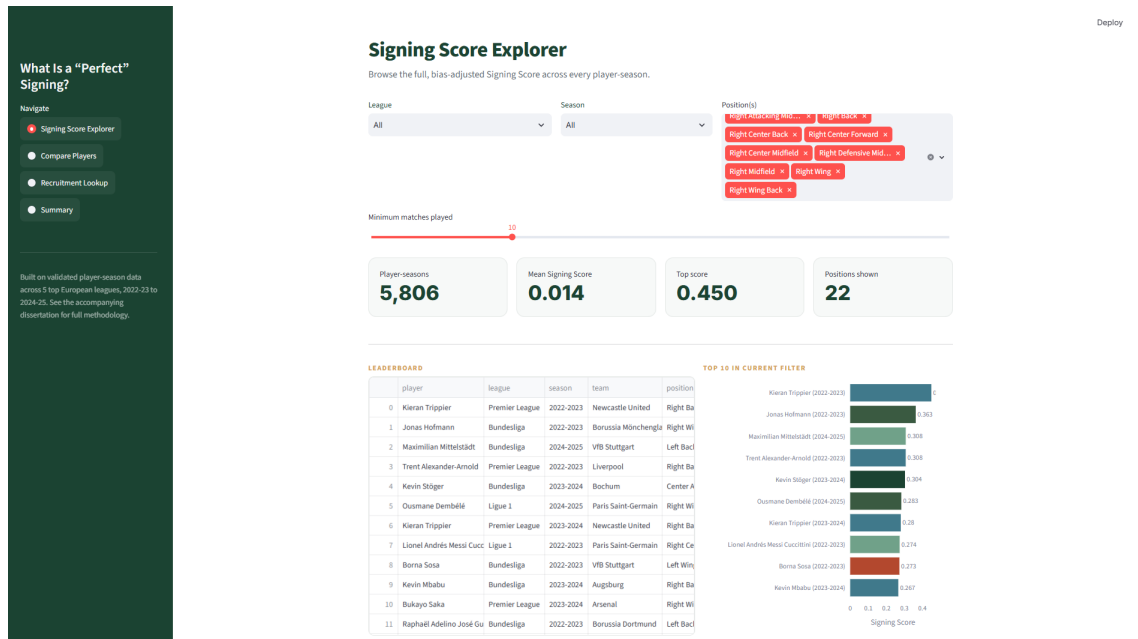


Fig. 23: Signing Score Explorer view refiltered to eight wide and defensive positions, illustrating the summary tiles updating to a substantially larger player-season count.

across all 742 qualifying transfers and cross-tabulating pre-move tier (top-half or bottom-half) against post-move flop status produces the 2×2 contingency table underlying the reported $\chi^2 = 69.16$ statistic (Section IX-A): a player in this example’s position (top-half pre-move) falls into the group that historically flopped only 11.6% of the time, illustrating concretely what the aggregate statistic means for a single, individual recruitment decision.

S. Threats to Validity

We organise remaining validity considerations using the standard internal/external/ construct validity framework. *Internal validity*, whether observed effects reflect the causal or associative claims made, is strengthened throughout by the discovery/confirmation design and, for two borderline findings, by repeated sub-sampling validation, but remains bounded by the single-split nature of the original design for the four myths not subjected to repeated validation, and by the correlational (not experimental) nature of every test in this paper: no random assignment of players to teams, ages, or transfer events is possible in observational football data, and all causal language in this paper (e.g. regarding the New Manager Bounce or Adaptation Tax findings) should be read as “causally-informed observational inference” via matching and baseline comparison, not as evidence from a randomised experiment. *External validity* (whether findings generalise beyond the specific sample studied) is supported by the five-league, three-season scope of the dataset, but is explicitly bounded by the confirmed Premier-League-specific anomaly (Section V-E), which indicates that at least one relationship studied in this paper (OBV’s relationship to team success) does not generalise uniformly across the leagues in scope, and should caution against assuming other findings

generalise perfectly across leagues either, even where not directly tested. *Construct validity*, whether OBV and its derived Signing Score actually measure the underlying construct of interest (footballing quality relevant to a signing decision), rests primarily on the xT cross-validation (Section V-B) and the external transfer-outcome evaluation (Section IX-A); the latter is, in our view, the stronger of the two forms of evidence, since it tests the construct’s relationship to a real-world outcome of direct practical interest, rather than agreement with a second, similarly-constructed proxy metric.

T. Reproducibility Statement

All analysis is implemented as a sequence of seven Jupyter notebooks corresponding directly to the structure of this paper: data foundation and preparation (Section III); value framework construction and validation (Section V); myth-busting, including the repeated sub-sampling extension (Section VI); Signing Score construction, including the hierarchical Bayesian extension attempt (Section VII); external evaluation (Section IX); and a dedicated hierarchical Bayesian refinement notebook and a dedicated repeated random sub-sampling robustness-check notebook, both constructed as isolated extensions to the core pipeline to avoid risk to already-validated upstream results. Intermediate results at each pipeline stage are checkpointed to disk and independently re-verified (row counts, duplicate checks, and reload consistency) before being consumed by downstream stages, a discipline applied consistently across all seven notebooks and referenced throughout this paper wherever a specific verification step is described.



Fig. 24: Leaderboard, top-10 bar chart, and multi-position score-distribution histogram for the wide/defensive filter of Figure 23.

U. Computational Environment

All analysis is implemented in Python. Core data manipulation uses pandas [24] and pyarrow (Parquet-format checkpointing between pipeline stages); statistical testing uses scipy.stats [25] for classical hypothesis tests and correlation, and statsmodels [26] for linear regression. The hierarchical Bayesian extension (Section VII-E) uses PyMC [6] with its default NUTS sampler [9] and ArviZ for convergence diagnostics [7], [21]. The interactive dashboard (Section X) is implemented in Streamlit, reading directly from the project’s validated Parquet checkpoint files at runtime.

V. Supplementary Figures

W. Notation and Glossary

The mathematical symbols used throughout this paper are defined at first use and collected for reference in Table I (Section II). Table VIII below additionally collects the non-mathematical abbreviations used across the paper, which are not all gathered in one place elsewhere.

TABLE VIII: Glossary of abbreviations

Abbreviation	Meaning
xT	Expected Threat
VAEP	Valuing Actions by Estimating Probabilities
EPV	Expected Possession Value
OBV	On-Ball Value
RAE	Relative Age Effect
FDR	False Discovery Rate
NUTS	No-U-Turn Sampler
MCMC	Markov Chain Monte Carlo
BDA	Bayesian Data Analysis
ppg	Points per game
NDA	Non-Disclosure Agreement

Two symbols in Table I are extended slightly for the hierarchical-model discussion in Appendix X: μ denotes the population-level mean and τ the population-level scale (between-group variance) in the hierarchical partial-pooling model of Section VII-E, with θ_p denoting the resulting position-group-level mean.

X. Formalised Procedures: Shrinkage Selection and Hierarchical Fitting

Sections VII-D and VII-E describe, in narrative form, the procedure used to select the Signing Score’s shrinkage constant and the protocol followed when fitting and evaluating the attempted hierarchical Bayesian extension. Both are formalised here as explicit algorithms, for readers who want the exact decision sequence rather than the prose description.

Algorithm 2 (Table IX) makes explicit that the shrinkage constant is not chosen by predictive validity alone: the grid search first identifies the empirically optimal candidate, and a separate behavioural constraint (the maximum share of a complete-season score attributable to the group prior rather than the player’s own evidence) is then used to override that raw recommendation, exactly as described in Section VII-D.

Applied with β set at the point where the group prior would outweigh a full season of individual evidence, this recovers $k_{\text{grid}} = 50$ at step 5 and $k^* = 10$ at step 7, matching the values reported in Section VII-D and Table VII.

Algorithm 3 (Table X) summarises the fitting and adopt/reject protocol applied to each of the two hierarchical model variants discussed in Section VII-E: a model is only adopted in place of the simpler global-constant formulation if it both converges cleanly and improves materially on the simpler alternative it would replace.

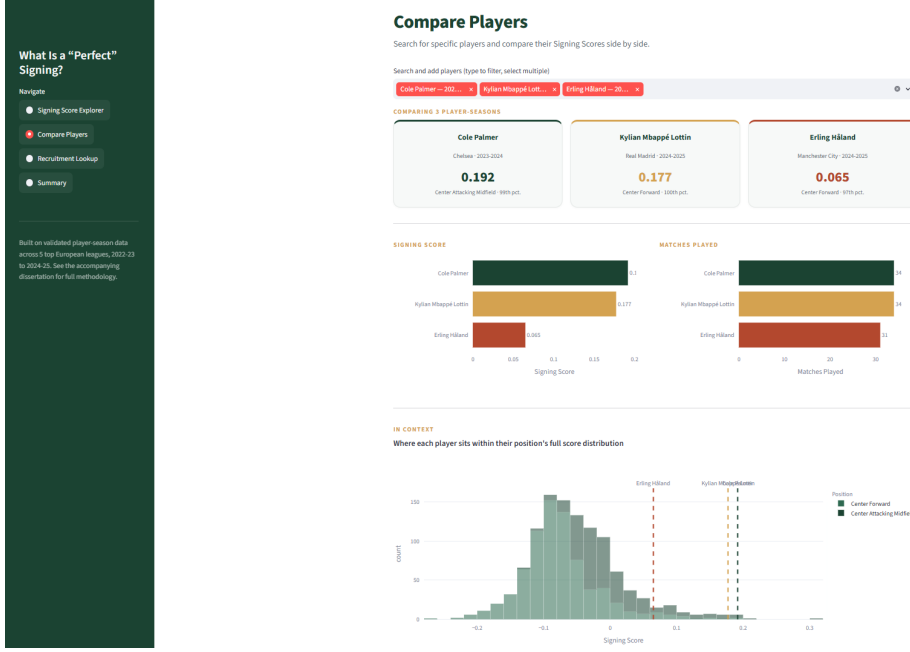


Fig. 25: Compare Players view: Cole Palmer, Kylian Mbappé, and Erling Haaland compared directly on Signing Score and matches played, with each player’s position located against the full population distribution.

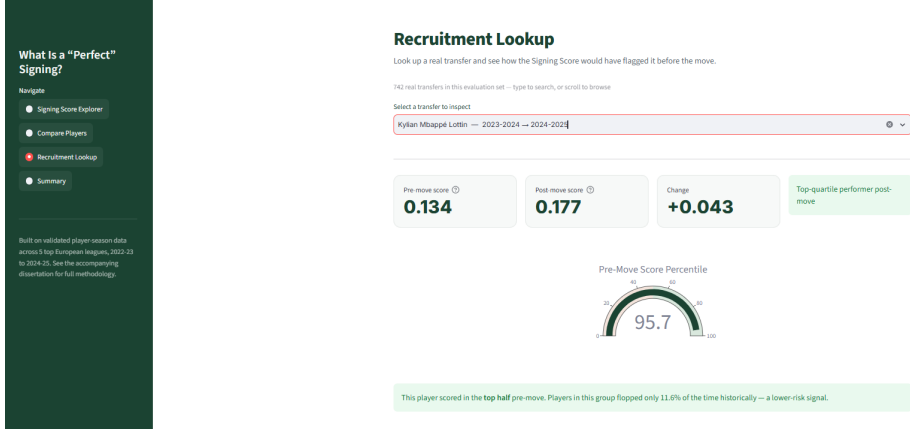


Fig. 26: Recruitment Lookup view applied to a real qualifying transfer (Kylian Mbappé, 2023–24 to 2024–25): pre-/post-move score, percentile gauge, and flop-rate context banner.

TABLE IX: Algorithm 2: Shrinkage constant selection

Input: candidate values $K = \{5, 10, 15, \dots, 200\}$; validation split; maximum matches played $n_{\max} = 38$; behavioural threshold β (maximum acceptable full-season group-mean share)

Output: selected shrinkage constant k^*

1. **for** each k in K :
2. Compute shrunk scores for all players via Equation 7
3. $r(k) \leftarrow$ out-of-sample predictive correlation against the validation target
4. $\text{share}(k) \leftarrow k / (n_{\max} + k)$
5. $k_{\text{grid}} \leftarrow \arg \max_k r(k)$
6. $K_{\text{admissible}} \leftarrow \{k \in K : \text{share}(k) \leq \beta\}$
7. $k^* \leftarrow \arg \max_{k \in K_{\text{admissible}}} r(k)$
8. **return** k^* , together with the full $(k, r(k))$ grid for transparency

both: the first converges cleanly at step 2 but fails the materiality check at step 7 (maximum difference from simple per-position averages of only 0.0025); the second never clears step 5, failing its reliability diagnostic across all three remediation attempts while its point estimate remains stable throughout, indicating a structural sampling difficulty specific to the model’s parameterisation rather than an under-explored posterior. Both outcomes are reported in full in Section VII-E and Appendix N, consistent with this paper’s stated discipline of reporting a validation failure rather than omitting it.

Y. Notebook and Pipeline Structure

As stated in the Reproducibility Statement (Appendix T), the analysis is organised as seven Jupyter notebooks plus a separate dashboard application, each corresponding to a

Applied to the two variants attempted, this protocol rejects

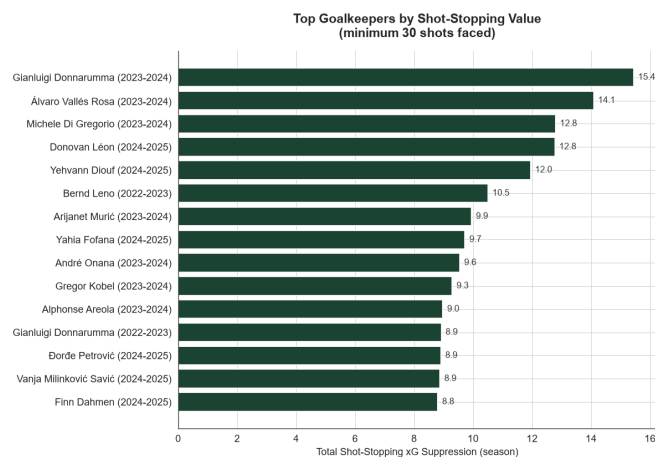


Fig. 27: Top goalkeepers by raw shot-stopping suppression, prior to Beta-Binomial shrinkage correction (Section V-F), included for comparison against the corrected leaderboard (Figure 12).

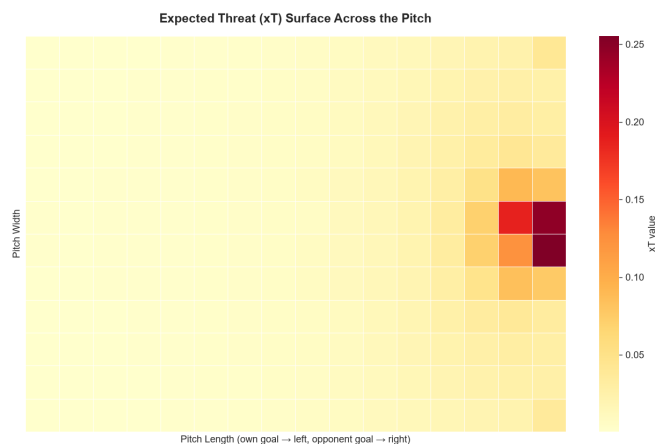


Fig. 28: Solved Expected Threat (xT) zone-value surface across the 16×12 pitch grid (Section V-B), showing the expected smooth increase in zone value approaching the opponent's goal.

specific section of this paper. Table XI summarises the primary inputs and outputs of each, to make the correspondence between the pipeline and the paper's structure explicit for anyone reproducing this work end-to-end.

Each notebook reads only the checkpointed Parquet outputs of its upstream dependencies rather than re-deriving them, and each independently re-verifies row counts, duplicate keys, and reload consistency on read, so that a failure introduced upstream surfaces at the point of consumption rather than silently propagating into a later section's results.

Expanding on Table XI, Notebook 1 is the widest-scoped of the seven: it is where the cross-provider identity-matching problem described in Section III is actually resolved, where the position-granularity data-quality issue (Section III-C) was first noticed and patched, and where the recurring cross-league merge pattern (Section III-G) is applied five times, once per league, before the resulting frames are concatenated

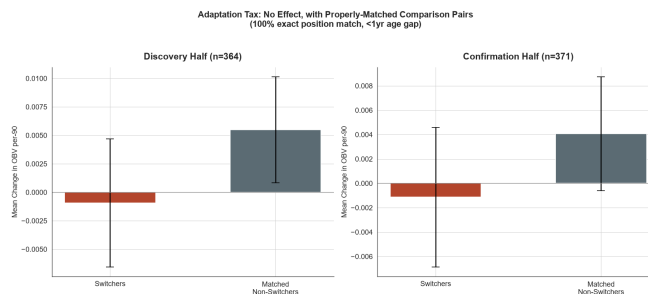


Fig. 29: Adaptation Tax: performance change for switching players versus the matched non-switching comparison population, following stratified matching correction (Section VI-B).

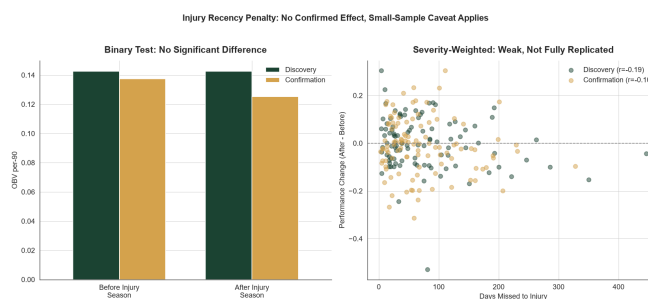


Fig. 30: Injury Recency Penalty: binary (injured versus not) and severity-weighted (days missed versus performance change) test results, both halves (Section VI-F).

and checkpointed. Because every other notebook depends on its output, its own internal checks are the strictest in the pipeline: row counts are compared against each raw source file immediately after every join, and any unexplained row-count drift halts execution rather than silently propagating a partial merge downstream.

Notebook 2 consumes Notebook 1's checkpoint and is where the value framework itself is built and stress-tested: the OBV coverage analysis, the independent xT cross-validation, the defensive-activity supplementary metric, the external validation against team success, the investigation of the Premier League anomaly, and the Beta-Binomial goalkeeper shrinkage all live here, together with the single-match SecondSpectrum tracking-data illustration used in Section V-G. It involves the tracking-data frame-sampling step described in Section V-G, which is more computationally intensive than the other pipeline stages.

Notebook 3 is comparatively self-contained: it reads only the checkpoints it needs for each of the six recruitment-belief tests, applies the Benjamini–Hochberg FDR correction [11] across the six simultaneously rather than testing them in isolation, and writes out the discovery/confirmation-split verdict table that Section VI reports directly. Notebook 4 involves the most rejected and revised work of the seven, reflecting the corrections documented in Section VII: the team-strength adjustment, the age-adjustment test that was tried and rejected, the full shrinkage grid search, and the attempted

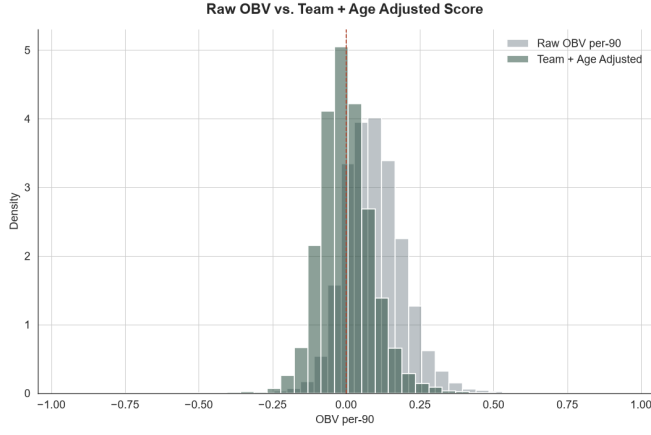


Fig. 31: Distribution of team-and-age-adjusted residual On-Ball Value, illustrating the combined effect of the team-strength correction prior to the age-adjustment test described in Section VII-C.

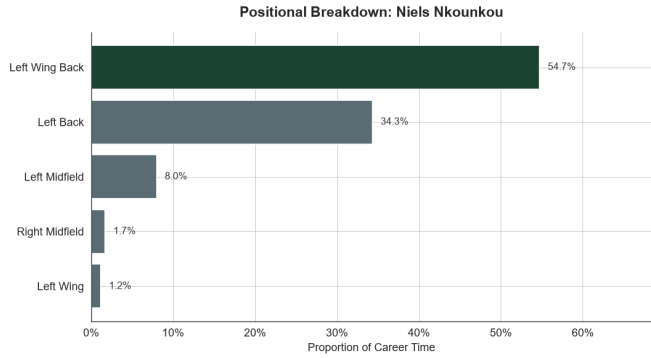


Fig. 32: A concrete individual example of positional versatility (Section III), grounding the population-level distribution shown in Figure 1 in one real, recognisable player-season case.

hierarchical Bayesian extension are each implemented as separable stages so that a rejected stage (age adjustment; the hierarchical extension) can be excluded from the production formula without disturbing the stages that were retained.

Notebook 5 is deliberately kept downstream of, and independent from, Notebooks 3 and 4: its transfer-outcome, durability, and team-aggregate evaluations are run against the already-finalised Signing Score formula, so that no evaluation choice in Notebook 5 could retroactively influence which correction was selected earlier in the pipeline. Notebooks 6 and 7 are the two extension notebooks referenced throughout this appendix: both were built after the core pipeline (Notebooks 1–5) was already validated and frozen, and both write their diagnostic output to separate checkpoint files so that a failed or inconclusive extension (as Notebook 6’s second model variant turned out to be) carries no risk of altering an already-validated upstream result.

1) *Code Repository*: The notebooks and dashboard application described above are version-controlled and made available at:

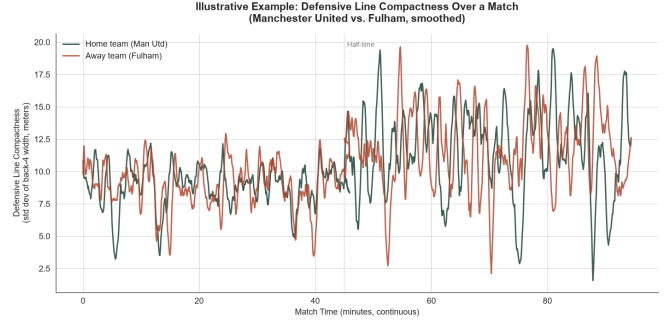


Fig. 33: Defensive line compactness over match time, single-match SecondSpectrum illustration, prior to correction of the half-time coordinate-flip artefact (Section V-G); included for direct comparison against the corrected version (Figure 7).

TABLE X: Algorithm 3: Hierarchical model fitting and adopt/reject protocol

Input: model variant M ; weakly-informative priors for μ , τ , $\{\theta_p\}$; maximum remediation attempts = 3
Output: adopt/reject decision; full diagnostic report
1. Fit M via PyMC’s NUTS sampler [6], [9] (4 chains)
2. Compute $\hat{R}$ [7] and effective sample size for every parameter via ArviZ [21]
3. if any $\hat{R} > 1.01$ or divergences > 0 :
4. for attempt = 1 to 3: apply a remediation (reparameterisation, higher target_accept, longer warm-up); re-fit; re-check
5. if still failing after 3 attempts: report unresolved; reject M
6. Compare M ’s position-level estimates to simple per-position averages (maximum absolute difference)
7. if diagnostics pass and improvement exceeds a pre-specified materiality threshold: adopt M
8. else : reject M ; retain the simpler formulation
9. return decision, full diagnostic report

[https:](https://github.com/GoswamiMeghana/signing-score-msc-project)

[//github.com/GoswamiMeghana/signing-score-msc-project](https://github.com/GoswamiMeghana/signing-score-msc-project)

As this project was undertaken under a company agreement with Nottingham Forest Football Club (Appendix Y3), the repository contains code only: the seven notebooks, the Streamlit dashboard application summarised in Table XI, a README describing the repository structure and setup instructions, and a `self_requirements.txt` listing the Python package versions used (pandas [24], pyarrow, scipy [25], statsmodels [26], PyMC [6], ArviZ [21], and Streamlit). No underlying data is included. All three licensed and company-provided data sources used in this project, StatsBomb event data, SecondSpectrum tracking data, and the injuries dataset, are subject to the data licence and NDA terms described in Appendix Y3 and are instead held in a private, access-controlled Wasabi object storage bucket, accessible only to the project team; none is ever committed to the repository or otherwise made public. The notebooks are configured to read directly from that protected storage location at runtime rather than from any file in the repository, so the repository itself contains only the analysis code and cannot be run end-to-end without separately-arranged access to that storage.

TABLE XI: Notebook-to-section correspondence

Notebook	Section(s)	Role
1	III	Data foundation: cross-provider identity matching, position/duration reconstruction, versatility scoring, cross-league merge, Parquet checkpointing.
2	V	Value framework: OBV construction, xT cross-validation, defensive-activity metric, team-success validation, Premier League anomaly, goalkeeper shrinkage.
3	VI	Mythbusting: the six recruitment-belief tests, FDR correction across the six, discovery/confirmation splitting.
4	VII	Signing Score: team-strength adjustment, age-adjustment test (rejected), shrinkage grid search and selection, hierarchical extension attempt, final formula.
5	IX	External evaluation: transfer-outcome, durability, and team-aggregate evaluation.
6 (ext.)	VII-E	Isolated hierarchical Bayesian refinement: full diagnostic suite for both model variants, run independently to avoid risk to the validated core pipeline.
7 (ext.)	VI, App. A	Isolated repeated random sub-sampling robustness check (Algorithm 1), applied to the two borderline mythbusting findings.
Dashboard	VIII, App. P	Streamlit application reading directly from the validated Parquet checkpoints; not a notebook, run as a standalone process.

To reproduce the pipeline: clone the repository, install the dependencies from `self_requirements.txt` into a fresh virtual environment, and run the seven notebooks in order (`_1Data_Foundation.ipynb` through `_7robustness_check.ipynb`), since each notebook consumes the checkpointed Parquet output of the ones before it; `_6Hierarchical_Bayesian_scoring_refinement.ipynb` and `_7robustness_check.ipynb` are independent extensions and may be run at any point after `_4Bias_Adjusted_Scoring.ipynb` and `_3Mythbusting.ipynb` respectively. The dashboard is launched separately, after the core pipeline has produced its checkpoint files, via `streamlit run self_app.py`. Exact commands are given in the repository’s README.

2) *Hyperparameter and Configuration Reference*: Table XII consolidates the fixed configuration constants referenced individually throughout the paper, gathered here as a single reference point for anyone reproducing a specific stage of the pipeline.

Several of these constants, α and the FDR method in particular, are applied uniformly across the paper precisely so that no single test’s threshold could be selectively loosened after the fact; the remainder are specific to the one stage of the pipeline noted alongside them in the table.

TABLE XII: Fixed configuration constants used throughout the pipeline

Constant	Value and where it is used
α	0.05; significance threshold for every classical hypothesis test in the paper.
k	10; selected outfield shrinkage constant (Section VII-D), chosen over the raw-optimal $k = 50$ on behavioural grounds.
R	100; number of repeated random sub-sampling partitions in Algorithm 1 (Section IV-B), applied to the two borderline mythbusting findings.
FDR method	Benjamini–Hochberg [11]; applied jointly across the eleven formal tests underlying the six myths of Section VI (Section VI-G).
NUTS chains	4; used for both hierarchical model variants in Section VII-E.
$\hat{R}$ threshold	1.01; the convergence-diagnostic cutoff applied in Algorithm 3, step 3.
Remediation attempts	3; maximum reparameterisation attempts before a non-converging model variant is reported as unresolved (Algorithm 3, step 4).
Tracking-data sampling	Every 125th frame; used for the single-match SecondSpectrum illustration of Section V-G.

3) *Ethics and Data Governance*: This project was undertaken as a company project in partnership with Nottingham Forest Football Club, and the data-governance considerations that follow reflect that arrangement. All player-level data used in the value-framework and mythbusting analyses (Sections V and VI) is drawn from vendor-licensed or already-public sources: StatsBomb event data ([1] describes the underlying event-data schema) was obtained under a data licence associated with this company partnership rather than StatsBomb’s separate free open-data programme, while the multi-league results and transfer records are drawn from already-public performance outcomes. None of this reflects any personal, medical, or contractual information about individual players. The SecondSpectrum tracking data used for the single-match illustration in Section V-G and the injuries dataset described in Section III-E were both provided under the terms of the same company project agreement and are used here only in the aggregate or single-match illustrative form already presented in this paper. None of the three company-provided or licensed sources, StatsBomb event data, SecondSpectrum tracking data, or the injuries dataset, is included in, or reproducible from, the project’s public code repository (Appendix Y), which contains code only.

The interactive dashboard described in Section X and Appendix P was developed for, and is deployed within, Nottingham Forest Football Club’s own recruitment workflow. As noted in Section X, the live deployment cannot be shared publicly under the terms of that agreement; the representative screenshots and walkthrough provided in Appendix P are included instead, with all player names and identifying values retained only where the same information is already public elsewhere (e.g. via publicly reported transfer activity), and never surfaced for a purpose beyond the illustration of the tool’s functionality. No aspect of the statistical findings reported in Sections V through IX depends

on information that is not derived from the public or vendor-licensed sources listed in Section III-A; the confidentiality constraints described here apply to the deployed tool and the two company-provided datasets specifically, not to the paper’s core analytical results.

Z. Consolidated Results Reference

The mythbusting verdicts are already gathered in Table II (Section VI). For convenience, Table XIII extends the same reference-table treatment to the paper’s other principal quantitative results, drawing together numbers that are each individually reported once, in the section noted, but that are otherwise scattered across the paper’s narrative.

TABLE XIII: Principal quantitative results outside the mythbusting chapter

Result	Value (Section)
xT cross-validation	$r = 0.768$ against independently constructed xT model (V-B)
OBV defensive coverage	Confirmed 41% coverage of goalkeeper-specific events motivating the dedicated goalkeeper track (V-F)
Team-success validation, raw OBV	$r = -0.721$ per-contributor (V-D)
Premier League anomaly	6 of 8 largest under-performers are Premier League teams (V-E)
Goalkeeper Beta-Binomial parameters	$\alpha \approx 21.42$, $\beta \approx 10.29$; ≈ 31.7 effective pseudo-shots (V-F)
Outfield shrinkage selection	$k = 10$ selected over raw-optimal $k = 50$; $r = 0.526$ out-of-sample (VII-D)
Hierarchical extension, variant 1	All $\hat{R} = 1.00$, zero divergences; max. difference from simple averages 0.0025 (VII-E)
Hierarchical extension, variant 2	$\hat{R} = 1.02$ – 1.04 unresolved across 3 remediation attempts; point estimate stable at ≈ 0.136 (VII-E)
Transfer-outcome evaluation	$r = 0.526$ pre/post-move; flop rate 11.6% vs. 38.3%, $\chi^2 = 69.16$, $p < 0.0001$, $n = 742$ (IX-A)
Durability evaluation	Higher score predicts higher subsequent injury incidence ($d = 0.423$, $p < 0.0001$), attributed to an exposure effect; severity not predicted ($r = -0.094$, $p = 0.117$) (IX-B)
Team-aggregate evaluation	Squad Signing Score vs. final rank $r = -0.101$, versus $r = -0.486$ for raw squad OBV (IX-C)

As a further worked illustration, applying the goalkeeper Beta-Binomial shrinkage formula (Section V-F) to a hypothetical goalkeeper who has saved 14 of 20 shots faced (a raw save rate of 0.700, close to but above the population average $\bar{p} \approx 0.6755$):

$$\text{Shrunk save rate} = \frac{14 + 21.42}{20 + 21.42 + 10.29} = \frac{35.42}{51.71} \approx 0.685 \quad (20)$$

The shrunk estimate (0.685) sits much closer to the population average (0.6755) than to the raw observed rate (0.700), correctly reflecting that 20 shots faced is a small sample relative to the $\alpha + \beta \approx 31.7$ pseudo-shots of population evidence built into the formula; a goalkeeper facing several hundred shots across a season would, by contrast, see their shrunk estimate sit close to their own raw rate, since their

own evidence would then substantially outweigh the fixed population-derived prior. This is the same behaviour, driven by the same underlying logic, as the outfield worked example in Appendix X, applied here to the goalkeeper-specific formula rather than the outfield shrinkage constant k .

Read together, Tables II and XIII give a single-page view of every principal quantitative claim made in this paper.